~Supplementary material~

PTMoreR: a motif-centric analysis enabling cross-species PTM mapping and comparative phosphoproteomics across mammals

Supplementary Notes

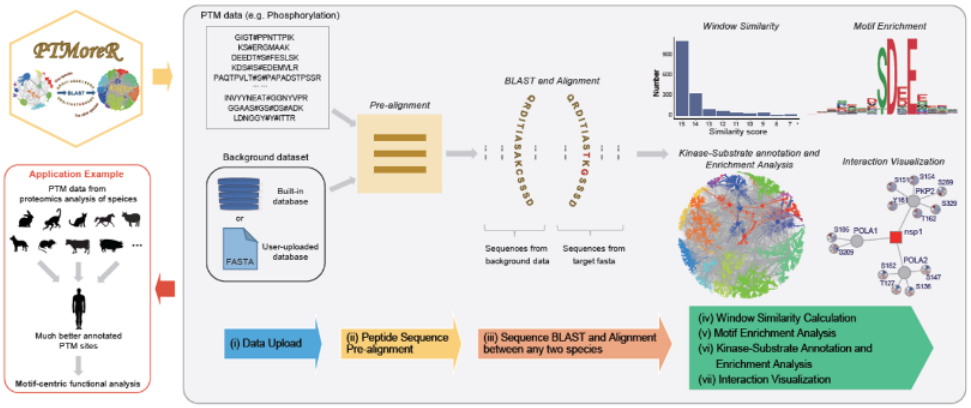
Overview *PTMoreR* (Post-translational modification ortholog aligner) is not merely a P-site BLAST tool; instead, it considers the surrounding amino acid sequence of PTM sites during BLAST, enabling a motif-centric analysis across species. Particularly, *PTMoreR* supports: 1. Mapping the PTM sites and protein sequences between any two species; 2. Calculating sequence window similarity and allowing filtering thresholds of sequence similarity during the mapping; 3. Processing PTM site-specific enrichment analysis and offering flexible annotation based on kinase-substrate database and network plots; 4. Visualizing the regulation of modification sites on the basis of protein-protein interaction data. Here we present the detailed introduction and operation of *PTMoreR*, by which users can follow to analyze their own data freely and conveniently.

Users can visit this site: https://yslproteomics.shinyapps.io/PTMoreR. The whole source codes: https://github.com/wangshisheng/PTMoreR. Then the website homepage can be shown like this:

PTMoreR

Welcome | Import Data | Pre-alignment | Blast | Motif Enrichment | KS Annotation | Interaction | Help

~~ Dear Users, Welcome to PTMoreR ~~



PTMoreR is a web-based tool, which possesses the core functions, including:

1. Mapping the PTM sites and protein sequences between any two species;
2. Calculating sequence window similarity and allowing filtering thresholds of sequence similarity during the mapping;
3. Performing PTM site-specific enrichment analysis and offering flexible annotations based on kinase-substrate database and network plots;
4. Visualizing the regulation of modification sites on the basis of protein-protein interaction data.

In addition, this tool supports both online access and local installation. The source codes and installation instructions can be available in the GitHub repository: https://github.com/wangshisheng/PTMoreR under an MIT license.

Finally, PTMoreR is developed by R shiny (Version 1.6.0), and is free and open to all users with no login requirement. It can be readily accessed by all popular web browsers including Google Chrome, Mozilla Firefox, Safari and Internet Explorer 10 (or later), and so on. We would highly appreciate that if you could send your feedback about any bug or feature request to Shisheng Wang at shishengwang@wchscu.cn.

Friendly suggestions:

a) Open PTMoreR with Chrome, Mozilla Firefox, Safari or Firefox;

b) The minimum operating system specifications are: RAM 4GB, Hard drive 100 GB;

c) The monitor resolution (>= 1920x1080) is better.

*^_^ Enjoy yourself in PTMoreR ^_^*

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1. Data Preparation

PTMoreR supports four basic file formats (.csv, .txt, .xlsx, .xls). Before analysis, users should prepare their peptide sequences with modification and protein background data. The modified peptide sequence data required here could be readily generated based on results of several popular tools such as MaxQuant (6), Spectronaut (7) and so on. The users then can upload the two data into *PTMoreR* with right formats respectively and start subsequent analysis.

1.1. Modified peptide sequences

1.1.1. Modified peptide sequences with normal type

In this situation, the first row is the column name (e.g. AnnotatedPeps) and each of the other rows is a modified peptide sequence. Users need to mark those modified residues (e.g. S, T, Y with phosphorylation) with some label they like (such as '#' or '@') in advance. The peptide sequences can be like:

AnnotatedPeps
GIGT#PPNTTPIK
GIGT#PPNT#T#PIK
NGS#PEIK
KS#ERGMAAK
MNGHS#DEESVR
RQIDS#S#EDEDDEDYDNDKR
RYS#GS#DS#DS#ISER
RY#S#GS#DS#DS#ISER
RY#S#GS#DS#DS#IS#ER
KRPY#S#S#FS#NGK

1.1.2. Modified peptide sequences from MaxQuant

If the sequence data are obtained from MaxQuant, then users can find the modified peptide sequences in the modification txt file, for example, the Phospho (STY)Sites.txt file in the output tables from MaxQuant. The peptide sequences are like this:

MaxQuant_Outputs
LFLDGEEEKEWAFEES(1)K
FDEGEDGEGS(0.996)NY(0.004)KKLC
ALVADEPEDLDT(1)EDEGLISFEEER
TYS(0.98)S(0.02)SGSSGGSHPSSR
ELILGS(0.002)ET(0.052)PS(0.779)S(0.167)PR
S(0.008)KS(0.992)PS(0.999)PPRLT(0.001)EDR
AAKLS(1)EGS(1)QPAEEEEDQETPSR
AAKLS(1)EGS(1)QPAEEEEDQETPSR
QEPT(1)QEHKQEEGQKQEEQEEEQEEEGK
NIGFKVNS(1)K

1.1.3. Modified peptide sequences from Spectronaut

If the sequence data are obtained from Spectronaut, then users can found the modified peptide sequences in the Standard Report part of Spectronaut, for example, export the Peptide Quant results from the Pivot Report and extract the modified peptide sequeces from the EG.ModifiedSequence column. The peptide sequences are like this:

Spectronaut_Outputs
INS[Phospho (STY)]APSS[Phospho (STY)]PIK
MLISAVS[Phospho (STY)]PEIR
KINS[Phospho (STY)]APSS[Phospho (STY)]PIK
KINS[Phospho (STY)]APS[Phospho (STY)]SPIK
INSAPSS[Phospho (STY)]PIK
EGSQGEPWT[Phospho (STY)]PTANLK
EGSQGEPWTPT[Phospho (STY)]ANLK
SHMSGS[Phospho (STY)]PGPGGSNTAPSTPVIGGSDKPGMEEK
SHMSGSPGPGGSNT[Phospho (STY)]APSTPVIGGSDKPGMEEK
SS[Phospho (STY)]SS[Phospho (STY)]LLASPGHISVK

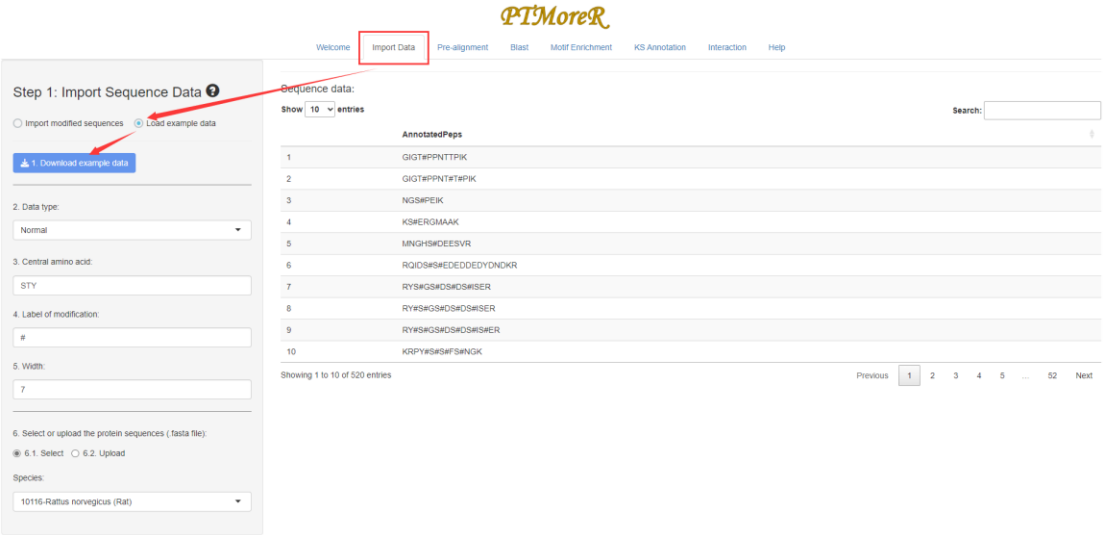
1.2. Background data

Background data here means the protein sequences of a species (.fasta format). Users should use the same protein sequence file as the background database. For example, users can download the protein sequences from UniProt (https://www.uniprot.org/) (3). The protein sequences like this:

```
>sp|Q64578|AT2A1_RAT Sarcoplasmic/endoplasmic reticulum calcium ATPase 1 OS=Rattus norvegicus OX=10116 GN=Atp2a1 PE=1 SV=1
MEAAHSKSTEECLSYFGVSETTGLTPDQVKRHLEKYGPNELPAEEGKSLWELVVEQFEDL
LVRILLLAACISFVLAWFEEGEETVTAFVEPFVILLILIANAIVGVWQERNAENAIEALK
EYEPEMGKVYRADRKSVQRIKARDIVPGDIVEVAVGDKVPADIRILSIKSTTLRVDQSIL
TGESVSVIKHTDPVPDPRAVNQDKKNMLFSGTNIAAGKAVGIVATTGVSTEIGKIRDQMA
ATEQDKTPLQQKLDEFGEQLSKVISLICVAVWLINIGHFNDPVHGGSWFRGAIYYFKIAV
ALAVAAIPEGLPAVITTCLALGTRRMAKKNAIVRSLPSVETLGCTSVICSDKTGTLTTNQ
MSVCKMFIIDKVDGDICSLNEFSITGSTYAPEGEVLKNDKPVRAGQYDGLVELATICALC
NDSSLDFNETKGVYEKVGEATETALTTLVEKMNVFNTEVRSLSKVERANACNSVIRQLMK
KEFTLEFSRDRKSMSVYCSPAKSSRAAVGNKMFVKGAPEGVIDRCNYVRVGTTRVPLTGP
VKEKIMSVIKEWGTGRDTLRCLALATRDTPPKREEMVLDDSAKFMEYEMDLTFVGVVGML
DPPRKEVTGSIQLCRDAGIRVIMITGDNKGTAIAICRRIGIFSENEEVADRAYTGREFDD
LPLAEQREACRRACCFARVEPSHKSKIVEYLQSYDEITAMTGDGVNDAPALKKAEIGIAM
GSGTAVAKTASEMVLADDNFSTIVAAVEEGRAIYNNMKQFIRYLISSNVGEVVCIFLTAA
LGLPEALIPVQLLWVNLVTDGLPATALGFNPPDLDIMDRPPRSPKEPLISGWLFFRYMAI
GGYVGAATVGAAAWWFLYAEDGPHVSYHQLTHFMQCTEHNPEFDGLDCEVFEAPEPMTMA
LSVLVTIEMCNALNSLSENQSLLRMPPWVNIWLLGSICLSMSLHFLILYVDPLPMIFKLR
ALDFTQWLMVLKISLPVIGLDELLKFIARNYLEG
>sp|Q64568|AT2B3_RAT Plasma membrane calcium-transporting ATPase 3 OS=Rattus norvegicus OX=10116 GN=Atp2b3 PE=1 SV=2
MGDMANSSIEFHPKPQQQREVPHVGGFGCTLAELRSLMELRGAEALQKIQEAYGDVSGLC
RRLKTSPTEGLADNTNDLEKRRQIYGQNFIPPKQPKTFLQLVWEALQDVTLIILEVAAIV
SLGLSFYAPPGEESEACGNVSGGAEDEGEAEAGWIEGAAILLSVICVVLVTAFNDWSKEK
QFRGLQSRIEQEQKFTVIRNGQLLQVPVAALVVGDIAQVKYGDLLPADGVLIQGNDLKID
ESSLTGESDHVRKSADKDPMLLSGTHVMEGSGRMVVTAVGVNSQTGIIFTLLGAGGEEEE
KKDKKGKQQDGAMESSQTKAKKQDGAVAMEMQPLKSAEGGEMEEREKKKANVPKKEKSVL
QGKLTKLAVQIGKAGLVMSAITVIILVLYFVIETFVVDGRVWLAECTPVYVQYFVKFFII
GVTVLVVAVPEGLPLAVTISLAYSVKKMMKDNNLVRHLDACETMGNATAICSDKTGTLTT
NRMTVVQSYLGDTHYKEIPAPSALTPKILDLLVHAISINSAYTTKILPPEKEGALPRQVG
NKTECALLGFILDLKRDFQPVREQIPEDQLYKVYTFNSVRKSMSTVIRMPDGGFRLFSKG
ASEILLKKCTNILNSNGELRGFRPRDRDDMVKKIIEPMACDGLRTICIAYRDFSAIQEPD
WDNENEVVGDLTCIAVVGIEDPVRPEVPEAIRKCQRAGITVRMVTGDNINTARAIAAKCG
IIQPGEDFLCLEGKEFNRRIRNEKGEIEQERLDKVWPKLRVLARSSPTDKHTLVKGIIDS
TTGEQRQVVAVTGDGTNDGPALKKADVGFAMGIAGTDVAKEASDIILTDDNFTSIVKAVM
WGRNVYDSISKFLQFQLTVNVVAVIVAFTGACITQDSPLKAVQMLWVNLIMDTFASLALA
TEPPTESLLLRKPYGRDKPLISRTMMKNILGHAVYQLTIIFTLLFVGELFFDIDSGRNAP
LHSPPSEHYTIIFNTFVMMQLFNEINARKIHGERNVFDGIFSNPIFCTIVLGTFGIQIVI
VQFGGKPFSCSPLSTEQWLWCLFVGVGELVWGQVIATIPTSQLKCLKEAGHGPGKDEMTD
EELAEGEEEIDHAERELRRGQILWFRGLNRIQTQMEVVSTFKRSGSFQGAVRRRSSVLSQ
LHDVTNLSTPTHIRVVKAFRSSLYEGLEKPESKSCIHNFMATPEFLINDYTHNIPLIDDT
DVDENEERLRAPPPPPPNQNNNAIDSGIYLTTHATKSATSSAFSSRPGSPLHSMETSL
```

1.3. Download example datasets

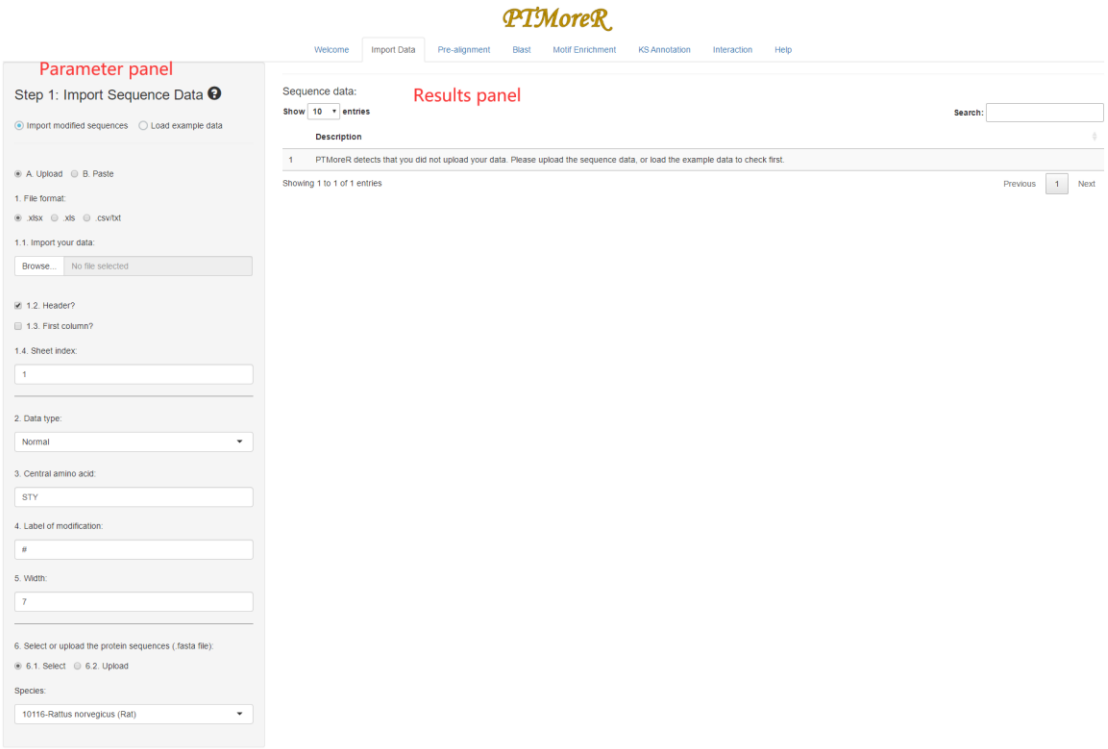
If users want to download the example datasets to their own computer and check the data format locally, they can download them from here:



First, select "*Load example data*" and the example data will be shown on the right panel interactively. Second, users can download the example data (the modified peptide sequences) with relative format (*2. Data type*: Normal, MaxQuant, Spectronaut) by clicking the corresponding button. The data sets are saved as .csv format and users can open them in other software, such as Excel.

2. Import Data

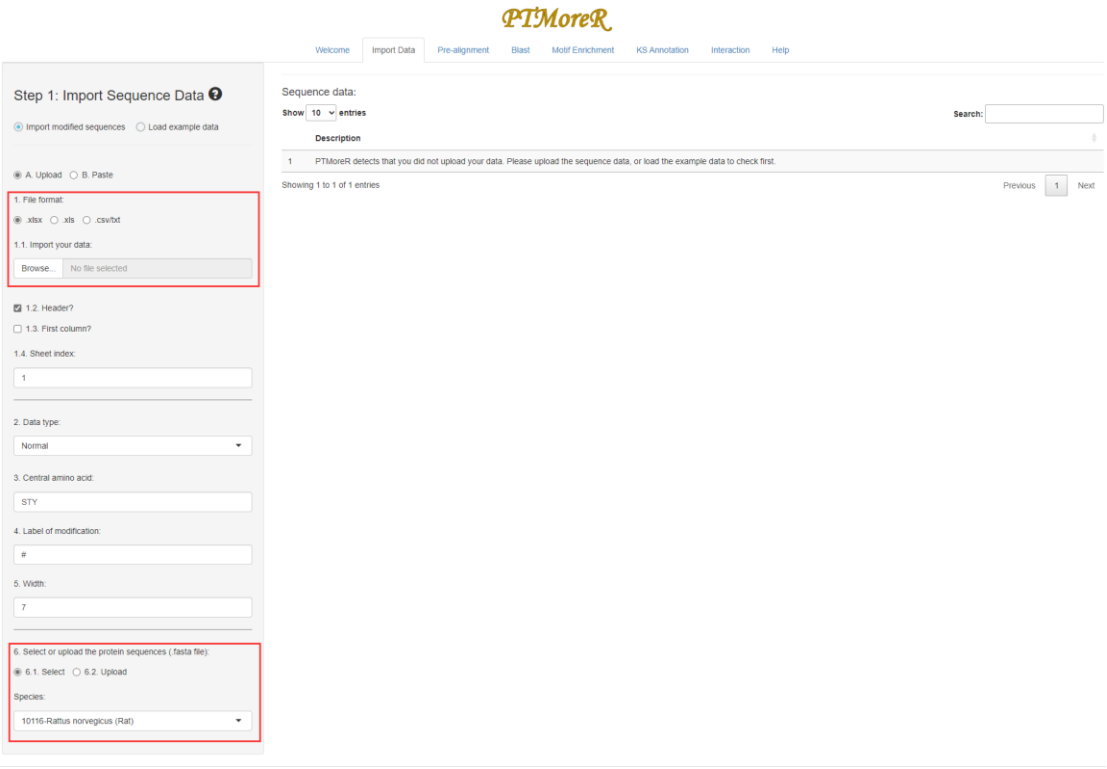
This is the first step, in which users should upload data here or load the example data with the above data formats. There are two main panels: first, *parameters panel*, users can adjust parameters here; second, *results panel*, many results after users set the parameters will be shown here and users can also download these results.



By default, we use the example data to show result of every step.

2.1. Uploading data.

When users prepare their data (the modified peptide sequences and protein background data), they can upload these data from here:



In the *parameters panel* of "*Step 1: Import Sequence Data*", there are two choices:

a. Import modified sequences. When users choose this option, they can choose "*A. Upload*" to upload their own data (the modified peptide sequences) here. Users should select the right format (*1. File format*: .csv, .txt, .xlsx, .xls) based on their data and then click "Browse" button (*1.1. Import your data*) to import the data;

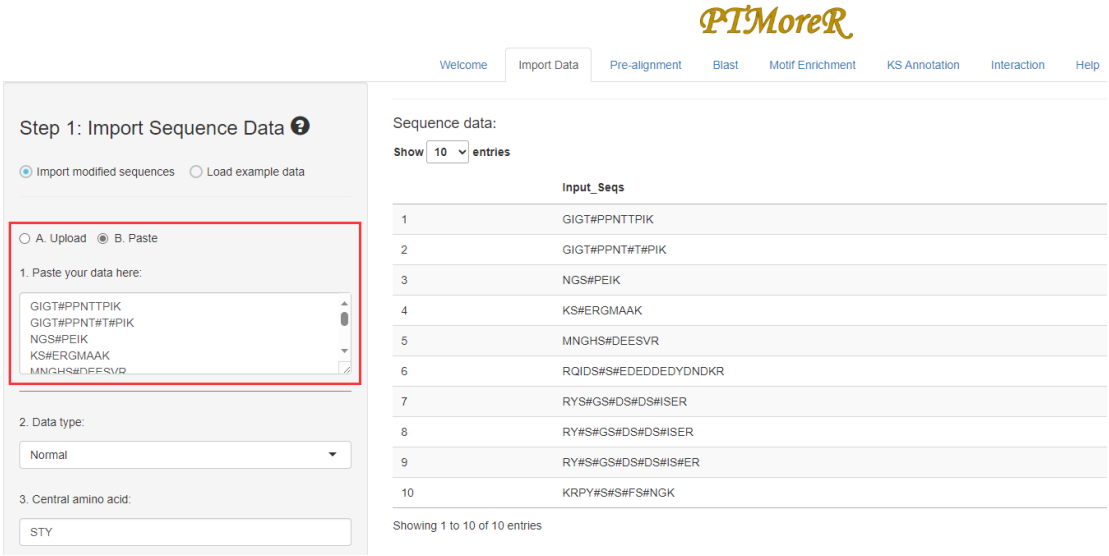
1.2. Header? this means whether the first row is column names. If true, you should choose this parameter.

1.3. First column: this means whether the first column is row names. If true, you should choose this parameter.

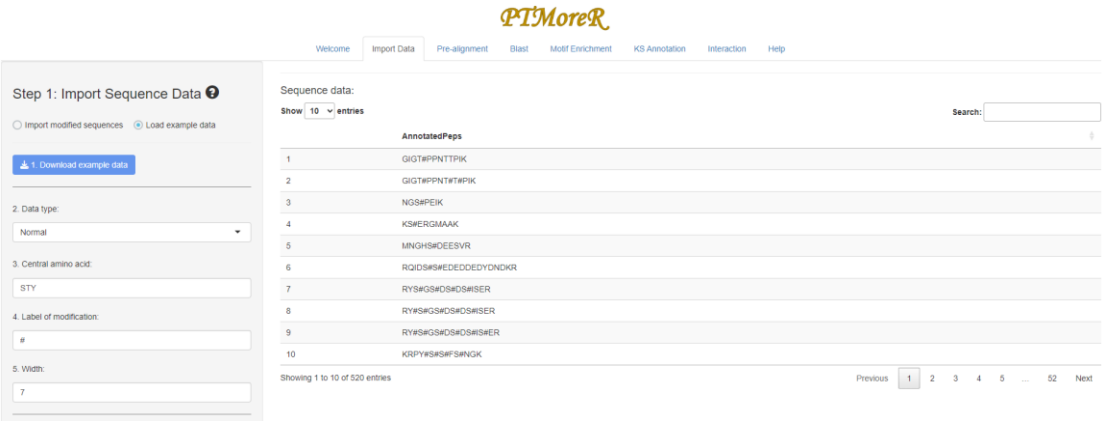
1.4. Sheet index: If users save their data into a .xlsx file, here means which sheet their data are.

In the *results panel* of "*Step 1: Import Sequence Data*", if users don't upload their data, here will show "*PTMoreR detects that you did not upload your data. Please upload the sequence data, or load the example data to check first.*" to warn users.

Users can also directly paste their sequences into *PTMoreR* by choosing "*B. Paste*" like below:

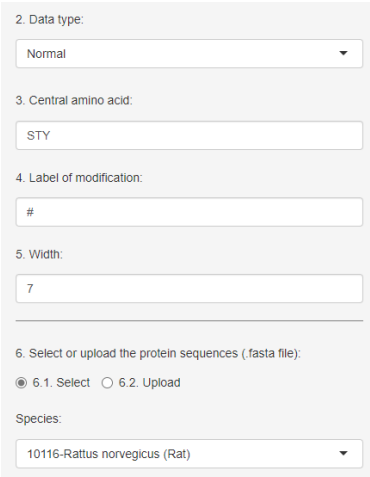


b. Load example data. As described in part 1.3 above, users can choose this option and download the example data to check them locally.



2.2. Parameters

There are some basic parameters that users can change based on their own data, shown as below:



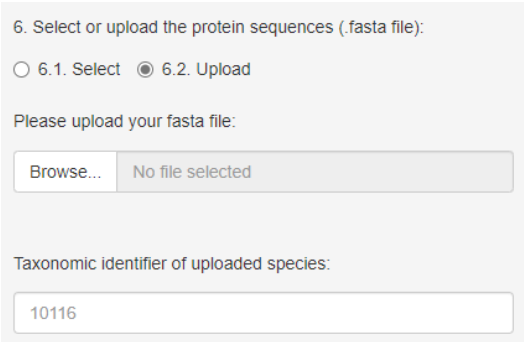
2. Data type: The original post-translational modification data obtained from which kind of search software. If you have processed the PTM data with standard format (e.g. NPT#Y#GSWFTEK), you should choose the "Normal", otherwise, if your PTM data are obtained from MaxQuant or Spectronaut, you should choose the relative type. Shown also as part 1.3 above.

3. Central amino acid: The central residue that users want to analyze, for example, phosphorylation motif analysis, can center on phosphorylated S, T or Y residues. If they want to analyze multi motif sites, here should be “STY”.

4. Label of modification: The label represents modification, users can use some label they like, such as “#”, “@”, in which “#” is recommended. Here is an example:

5. Width: It is the number of left/right side characters of the central residue.

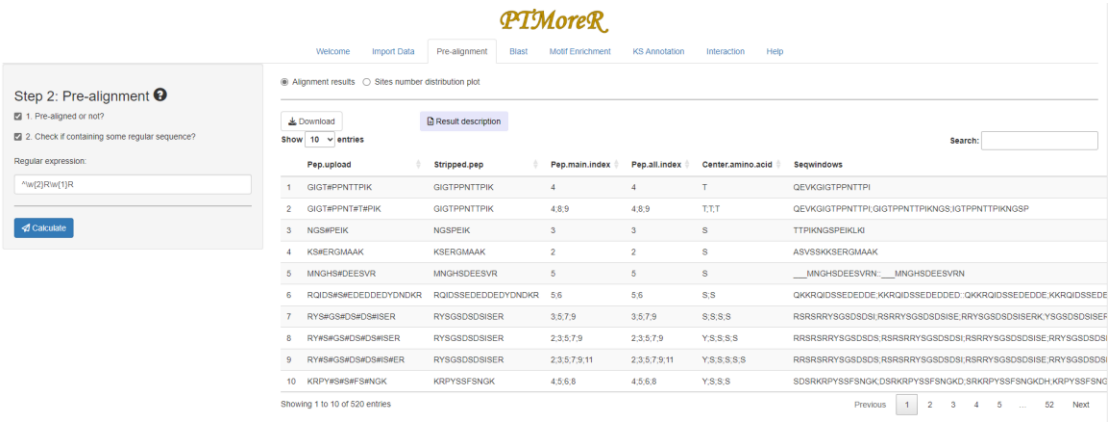
6. Select or upload the protein sequences (.fasta file): If users want to use the default database, they just select relative species, please note, the default database only have three species (Human, Mouse, Rat) with 15 length sequences now, more species data will be implemented in the future. Optionally, users can also upload their own .fasta file (*6.2. Upload*), no species limits here, but the calculation time would be longer.



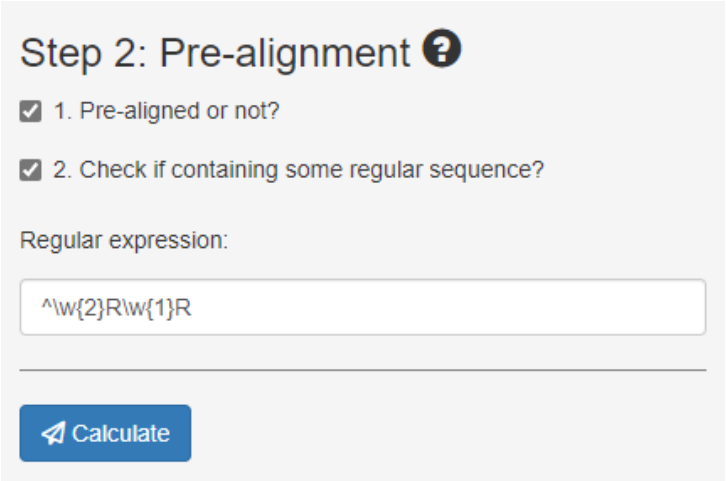
After uploading their own .fasta file, users should type in the taxonomic identifier, for example, rat's taxonomic identifier is 10116.

3. Pre-alignment

This step means align those peptide sequences with the background database (protein sequences) and force the modified sites/residues to be central sites, then users can get the standard peptide window sequences.



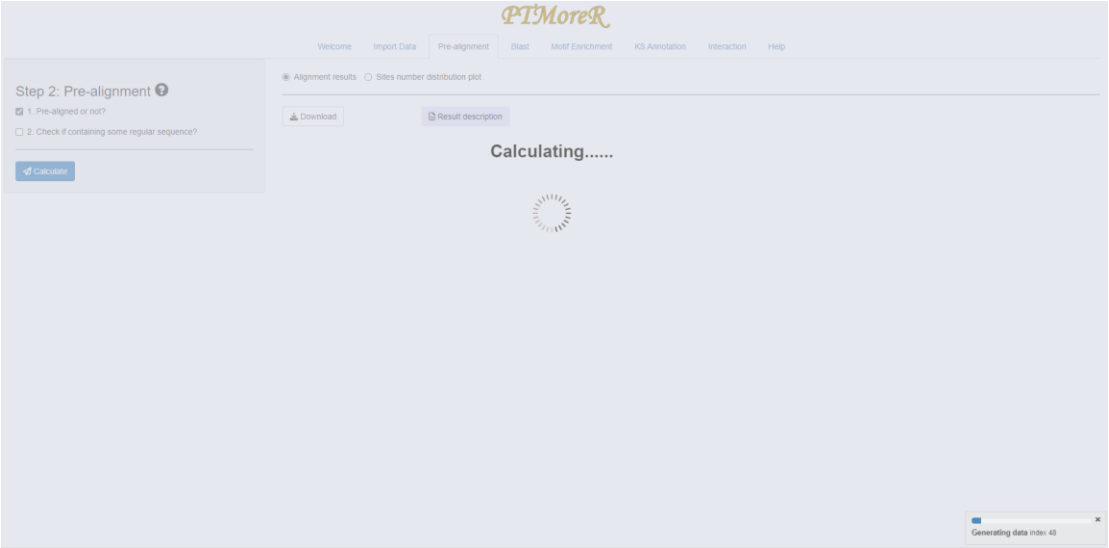
3.1. Parameters



1. Pre-aligned or not? here asks users whether pre-align their sequences, if your sequences are standard (e.g. 15 length amino acids), you can unselect this parameter. Default is true.

2. Check if containing some regular sequence: if users want to check whether the aligned peptides contain some specific sequences, for example, you want to find those peptides whose 3th and 5th position are R (arginine), then you can select this parameter and type in a simple regular expression, like "^\w{2}R\w{1}R" (more details can be found here: https://en.wikipedia.org/wiki/Regular_expression). Otherwise, you just unselect it.

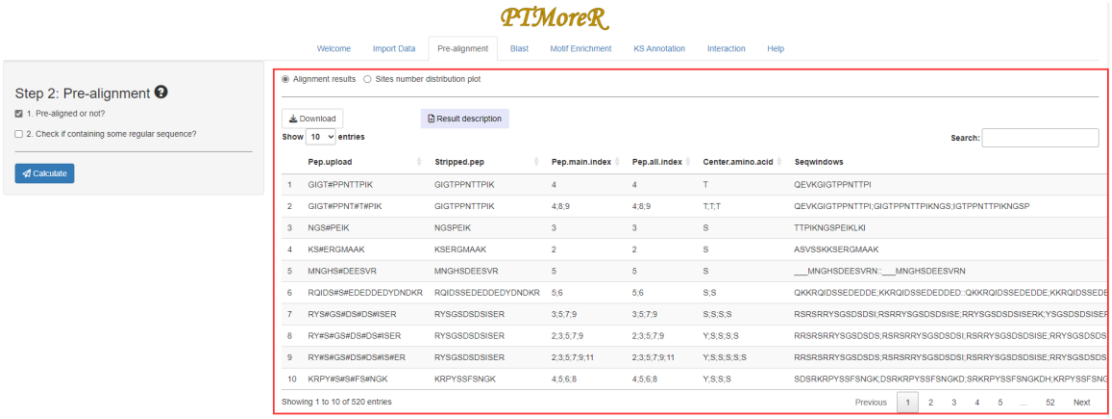
Then, you can click the “Calculate” button, it should be like this:



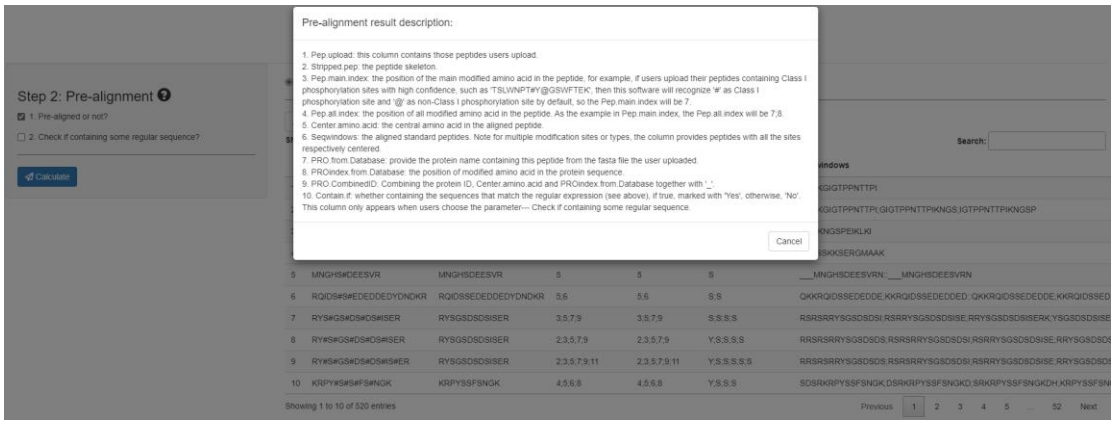
A process bar will appear in the bottom right corner to tell users where it goes.

3.2 Results

3.2.1. Alignment results



Users can click “Result description” button to read the introduction about every column, shown as below:



1. Pep.upload: This column contains those peptides users upload.

2. Stripped.pep: The peptide skeleton.

3. Pep.main.index: The position of the main modified amino acid in the peptide, for example, if users upload their peptides containing Class I phosphorylation sites with high confidence, such as 'TSLWNPT#Y@GSWFTEK', then this software will recognize '#' as Class I phosphorylation site and '@' as non-Class I phosphorylation site by default, so the Pep.main.index will be 7.

4. Pep.all.index: the position of all modified amino acid in the peptide. As the example in Pep.main.index, the Pep.all.index will be 7;8.

5. Center.amino.acid: the central amino acid in the aligned peptide.

6. Seqwindows: the aligned standard peptides. Note for multiple modification sites or types, the column provides peptides with all the sites respectively centered.

7. PRO.from.Database: provide the protein name containing this peptide from the fasta file the user uploaded.

8. PROindex.from.Database: the position of modified amino acid in the protein sequence.

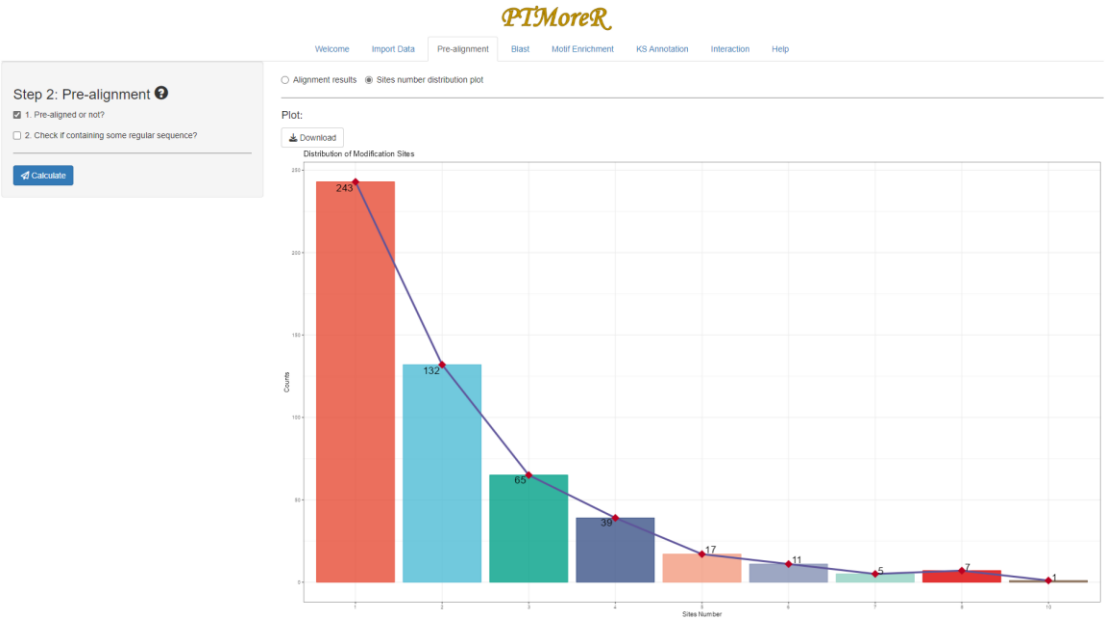
9. PRO.CombinedID: Combining the protein ID, Center.amino.acid and

PROindex.from.Database together with '_'.

10. Contain.if: whether containing the sequences that match the regular expression (see above), if true, marked with 'Yes', otherwise, 'No'. This column only appears when users choose the parameter--- Check if containing some regular sequence.

3.2.2. Sites number distribution plot

This software counts the number of modified sites and plot it:

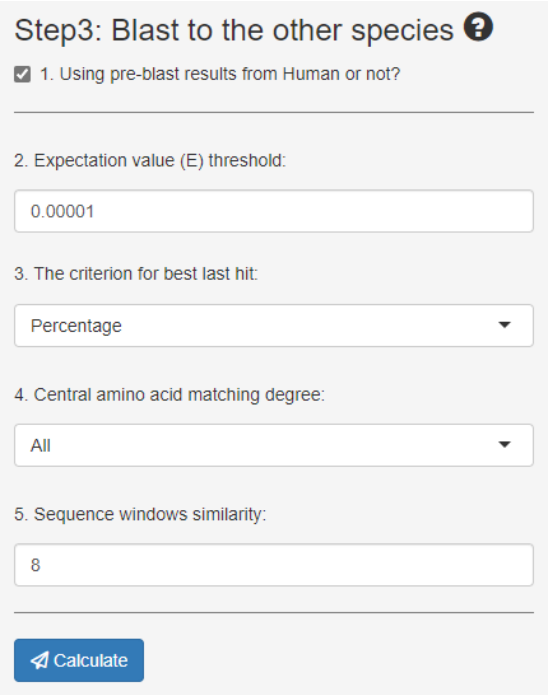


Users can also click “Download” button to save this plot as a .pdf file.

4. Blast to the other species

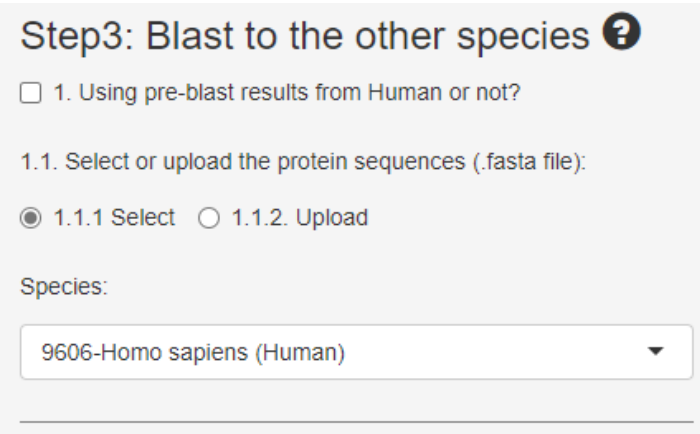
This step will map the PTM site and protein sequences and identifiers between the uploaded species and the other species.

4.1. Parameters

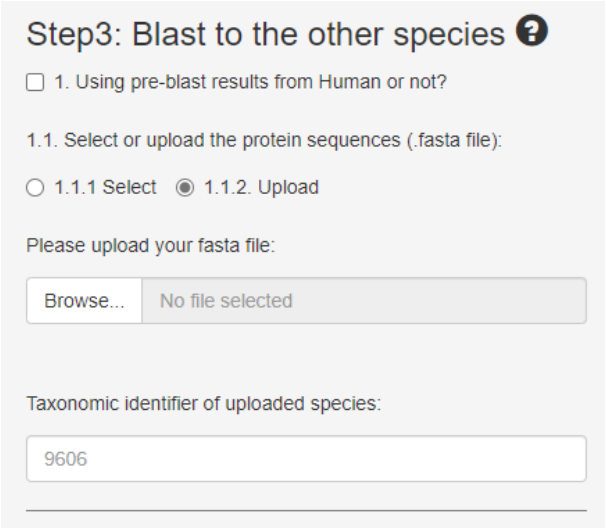


1. Using pre-blast results from Human or not? The pre-blast results here mean we have already blasted several species to Human and save the results in *PTMoreR*. If users use the pre-blast results, this tool will retrieve these results directly and it will be faster to finish this step. By default, it is true. Otherwise, if you want to blast other species, you can unselect this parameter and select/upload the .fasta file below. Please Note: This step is can be time-consuming! If users analyze the species that *PTMoreR* does not integrate, this tool will blast the uploaded sequences against those protein sequences of the other species, which will take more time, perhaps several hours.

Users can select a species that *PTMoreR* have integrated by selecting "*1.1.1. Select*":



If users want to analyze the species that *PTMoreR* does not integrate, they can choose "*1.1.2. Upload*" and upload the .fasta file and type in the taxonomic identifier, for example, human's taxonomic identifier is 9606.



2. Expectation value (E) threshold: Expectation value (E) threshold for saving hits.

3. The criterion for best last hit: This tool performs a BLAST search between query and subject sequences and returns only the best hit based on the selected criterion. "Percentage" means If e-values are identical then the hit with the largest matching percentage is chosen. "Longest alignment length" means If e-values are identical then the hit with the longest alignment length is chosen.

4. Central amino acid matching degree: The matching degree of central amino acids (CAAs) when the uploaded peptides are blasted to Human protein sequences. 1. Exact matching: The CAAs are same, for example, the CAA is "S" in the uploaded peptides and the CAA is also "S" in the blasted sequence. 2. Fuzzy matching: Only for phosphorylation, not for other modification type. For example, the CAA is "S" in the uploaded peptides and the CAA could be "S", "T", or "Y" in the blasted sequence. All: Reporting all results.

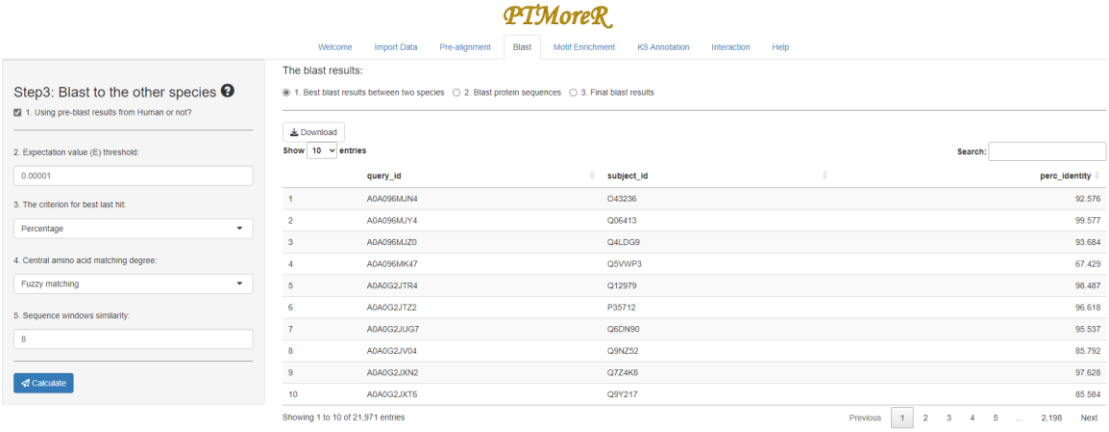
5. Sequence windows similarity: The similarity of sequence windows between the uploaded peptides and the blasted peptides. For example, there are 15 amino acids in one sequence window, 8 here means there are 8 amino acids are exactly same (amino acids names and positions in both windows are all the same).

4.2. Results

After setting proper parameters, users can click “Calculate” button and the results will be shown in the right panel.

4.2.1. Best blast results between two species

Here only shows the best hits between query and subject sequences, like below:



Three columns in this table:

query_id: The protein ids (e.g. UniProt IDs) that users want to query and select/upload in the Step 1 (2. Import Data part above).

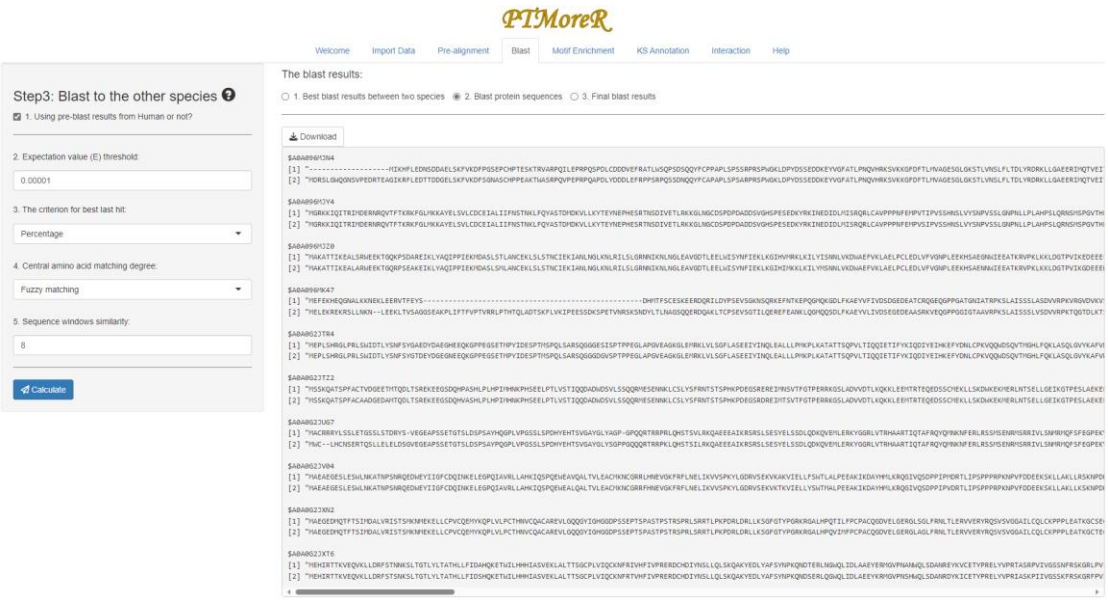
subject_id: The protein ids (e.g. UniProt IDs) that users want to map to and select/upload in this step.

perc_identity: Percent identity cutoff.

Users can click “Download” button to download the results, which are saved into a .RData file. RData is a kind of file format in R, which can be loaded easily using R.

4.2.2. Blast protein sequences

Here shows the mapping sequences between query and subject proteins.



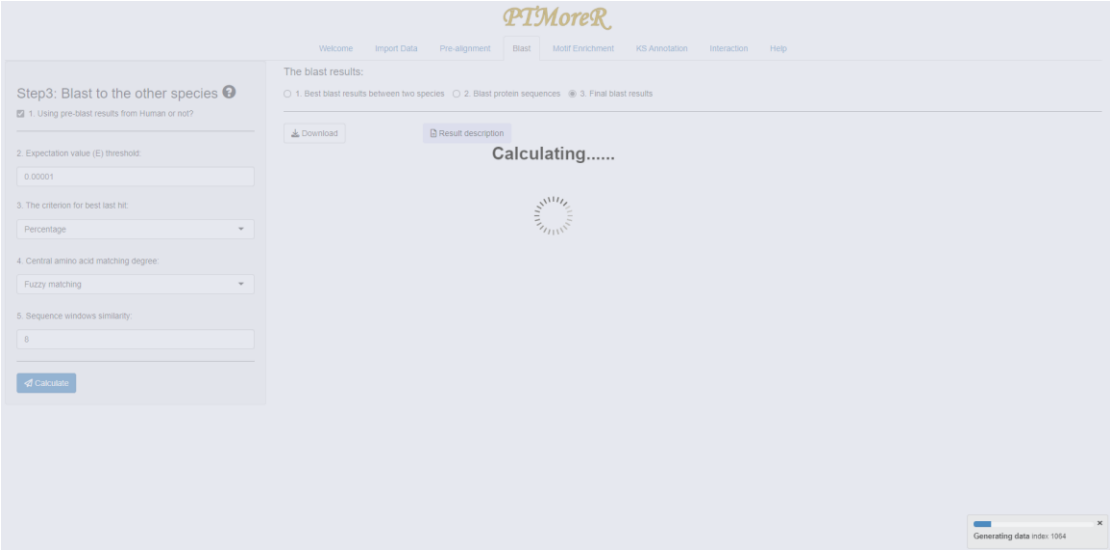
There are three rows for each protein. The first row is query protein id, the second row is the query protein sequence, the third row is the best blasted protein sequence.

```
$A0A096MJN4
[1] "-------------------MIKHFLEDNSDDAELSKFVKDFPGSEPCHPTESKTRVARPQILEP
[2] "MDRSLGWQGNSVPEDRTEAGIKRFLEDTTDDGELSKFVKDFSGNASCHPPEAKTWASRPQVPEP
```

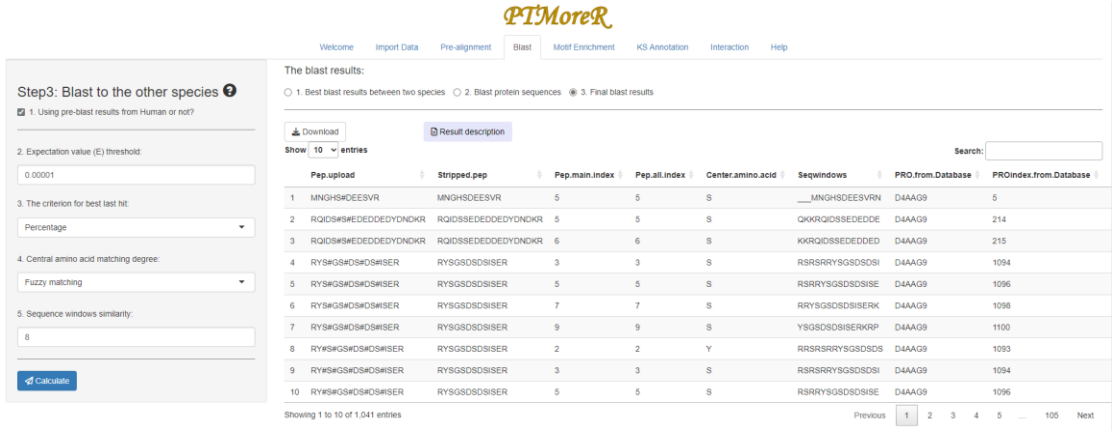
Users can also click “Download” button to download the results, which are saved into a .RData file.

4.2.3. Final blast results

When users click “3. Final blast results” here, this tool will calculate the final result, like below:



After that, users can obtain a table as below:



	Pep.upload	Stripped.pep	Pep.main.index	Pep.all.index	Center.amino.acid	Seqwindows	PRO.from.Database	PROindex.from.Database
1	MNGHS#DEESVR	MNGHSDEESVR	5	5	S	___MNGHSDEESVRN	D4AAG9	5
2	RQIDS#S#EDEDDEDYDNDKR	RQIDSSEDEDDEDYDNDKR	5	5	S	QKKRQIDSSEDEDDE	D4AAG9	214
3	RQIDS#S#EDEDDEDYDNDKR	RQIDSSEDEDDEDYDNDKR	6	6	S	KKRQIDSSEDEDDED	D4AAG9	215
4	RY$#GS#DS#DS#ISER	RYSGSDSDSISER	3	3	S	RSRSRRYSGSDSDSI	D4AAG9	1094
5	RY$#GS#DS#DS#ISER	RYSGSDSDSISER	5	5	S	RSRRYSGSDSDSISE	D4AAG9	1096
6	RY$#GS#DS#DS#ISER	RYSGSDSDSISER	7	7	S	RRYSGSDSDSISERK	D4AAG9	1098
7	RY$#GS#DS#DS#ISER	RYSGSDSDSISER	9	9	S	YSGSDSDSISERKRP	D4AAG9	1100
8	RY#S#GS#DS#DS#ISER	RYSGSDSDSISER	2	2	Y	RRSRSRRYSGSDSDS	D4AAG9	1093
9	RY#S#GS#DS#DS#ISER	RYSGSDSDSISER	3	3	S	RSRSRRYSGSDSDSI	D4AAG9	1094
10	RY#S#GS#DS#DS#ISER	RYSGSDSDSISER	5	5	S	RSRRYSGSDSDSISE	D4AAG9	1096

Users can click “Result description” button to read the introduction about every column, shown as below:

1. Pep.upload: this column contains those peptides users upload.

2. Stripped.pep: the peptide skeleton.

3. Pep.main.index: the position of the main modified amino acid in the peptide, for example, if users upload their peptides containing Class I phosphorylation sites with high confidence, such as 'TSLWNPT#Y@GSWFTEK', then this software will recognize '#' as Class I phosphorylation site and '@' as non-Class I phosphorylation site by default, so the Pep.main.index will be 7.

4. Pep.all.index: the position of all modified amino acid in the peptide. As the example in Pep.main.index, the Pep.all.index will be 7;8.

5. Center.amino.acid: the central amino acid in the aligned peptide.

6. Seqwindows: the aligned standard peptides.

7. PRO.from.Database: provide the protein ID/name containing this peptide from the fasta file the user uploaded.

8. PROindex.from.Database: the position of modified amino acid in the protein sequence.

9. PRO.CombinedID: Combining the protein ID, Center.amino.acid and PROindex.from.Database together with '_'.

10. Contain.if: whether containing the sequences that match the regular expression (see above), if true, marked with 'Yes', otherwise, 'No'. This column only appears when users choose the parameter---Check if containing some regular sequence (in step2).

11. Center.amino.acids.Other: the central amino acid mapped from the human peptides.

12. Seqwindows.Other: the standard peptides mapped from the human peptides.

13. PRO.from.Other: the protein ID/name from the mapped human protein.

14. PROindex.from.Other: the position of modified amino acid in the mapped human protein sequence.

15. PRO.CombinedID.Other: Combining the PRO.from.Other, Center.amino.acids.Other and PROindex.from.Other together with '_'.

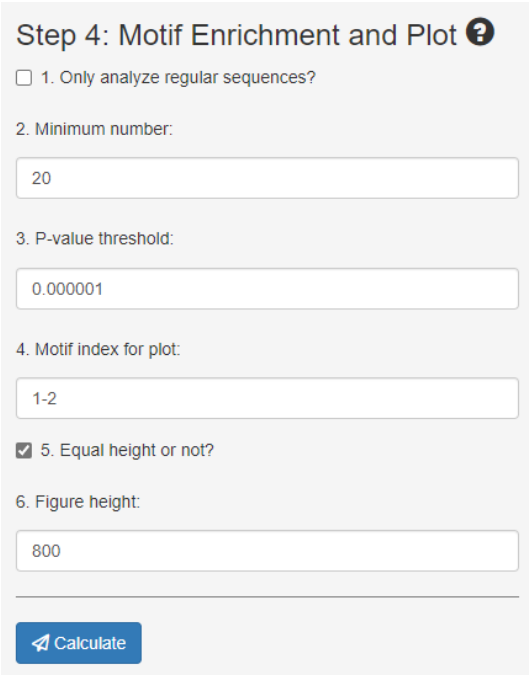
16. Center.aa.match: The matching degree of central amino acids (CAAs) when the uploaded peptides are blasted to Human protein sequences. 1. Exact matching: The CAAs are same, for example, the CAA is 'S' in the uploaded peptides and the CAA is also 'S' in the blasted sequence. 2. Fuzzy matching: For example, the CAA is 'S' in the uploaded peptides and the CAA could be 'S', 'T', or 'Y' in the blasted sequence. All: Reporting all results.

17. Window.similarity: The similarity of sequence windows between the uploaded peptides and the blasted peptides. For example, if there are 15 amino acids in one sequence window, 7 here means there are 7 amino acids are exactly same (amino acids names and positions in both windows are all same).

5. Motif Enrichment and Plot

This step will find overrepresented sequence motifs as we previous described for single species (8).

5.1. Parameters



1. Only analyze regular sequences? If true, this tool will only analyze the peptides contain some specific sequences (regular sequences) obtain from “3. Pre-alignment” part above.

2. Minimum number: This threshold refers to the minimum number of times you wish each of your extracted motifs to occur in the data set.

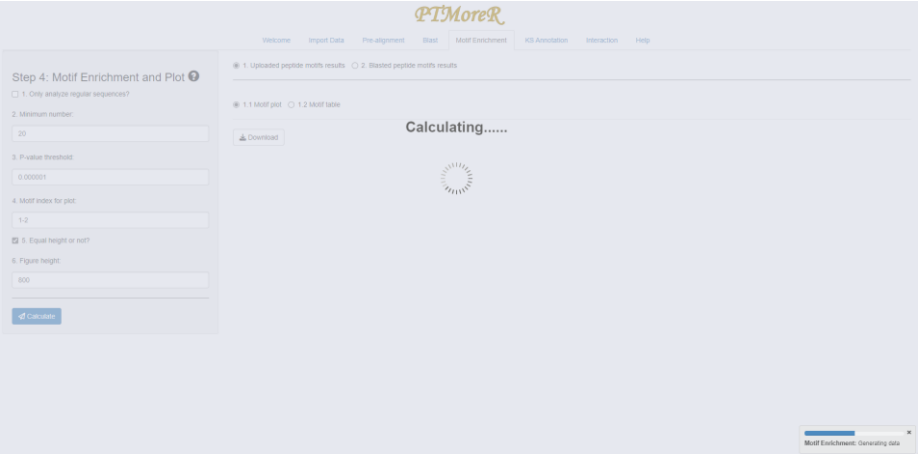
3. P-value threshold: The p-value threshold for the binomial probability. This is used for the selection of significant residue/position in the motif.

4. Motif index for plot: Which motif would be plotted. If users only type in one number, it will plot the relative motif. If users type in "1-10", it will plot the 1th to 10th motifs.

5. Equal height or not? Whether all residues in the figure have equal height. Default is false.

6. Figure height: The height of the figure.

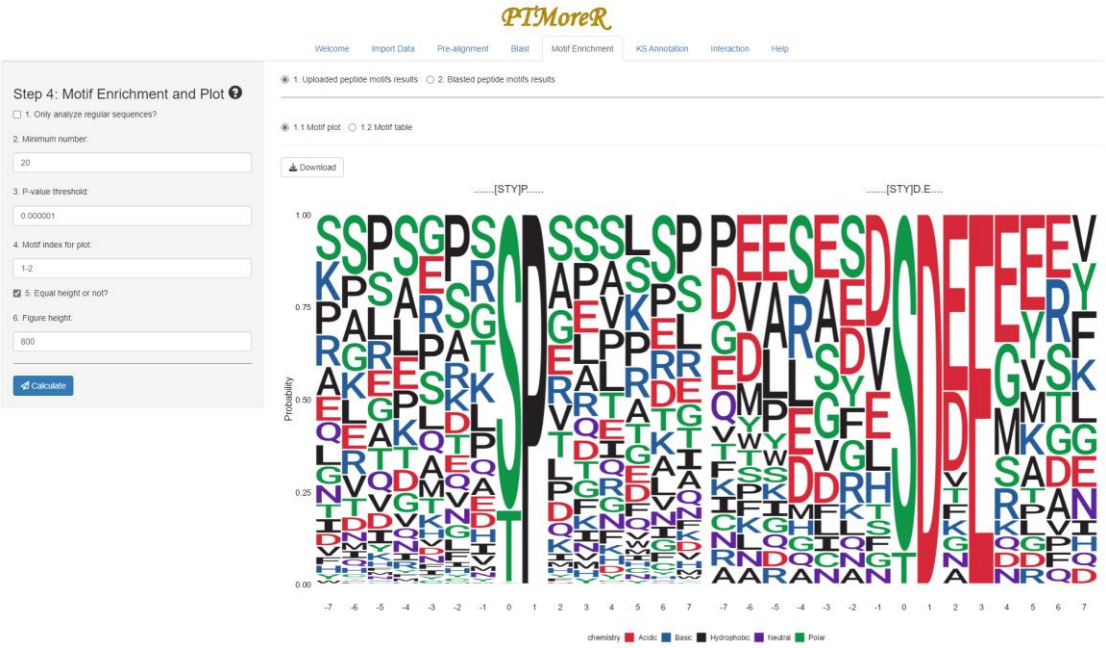
Then, users can click the “Calculate” button, this tool will process motif enrichment and plot those motifs:



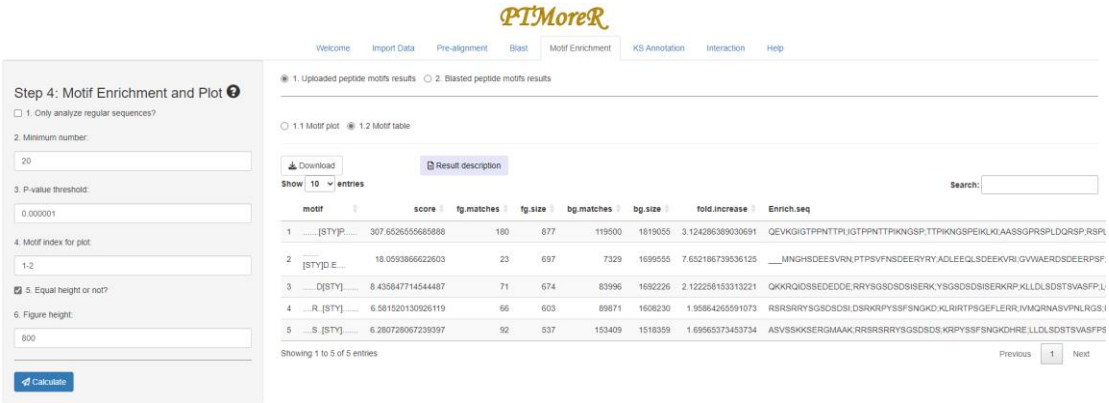
5.2. Results

5.2.1. Uploaded peptide motifs results

All results here are obtained from the uploaded peptides. When users click “1.1. Motif plot”, the motif plot is shown as below:



When users click “1.2. Motif table”, the motif table is shown as below:



Every column in this table means (Users can also check this by clicking “Result description” button):

1. motif: the overrepresented motif.

2. score: the motif score, which is calculated by taking the sum of the negative log probabilities used to fix each position of the motif. Higher motif scores typically correspond to motifs that are more statistically significant as well as more specific.

3. fg.matches: frequency of sequences matching this motif in the foreground set.

4. fg.size: total number of foreground sequences.

5. bg.matches: frequency of sequences matching this motif in the background set.

6. bg.size: total number of background sequences.

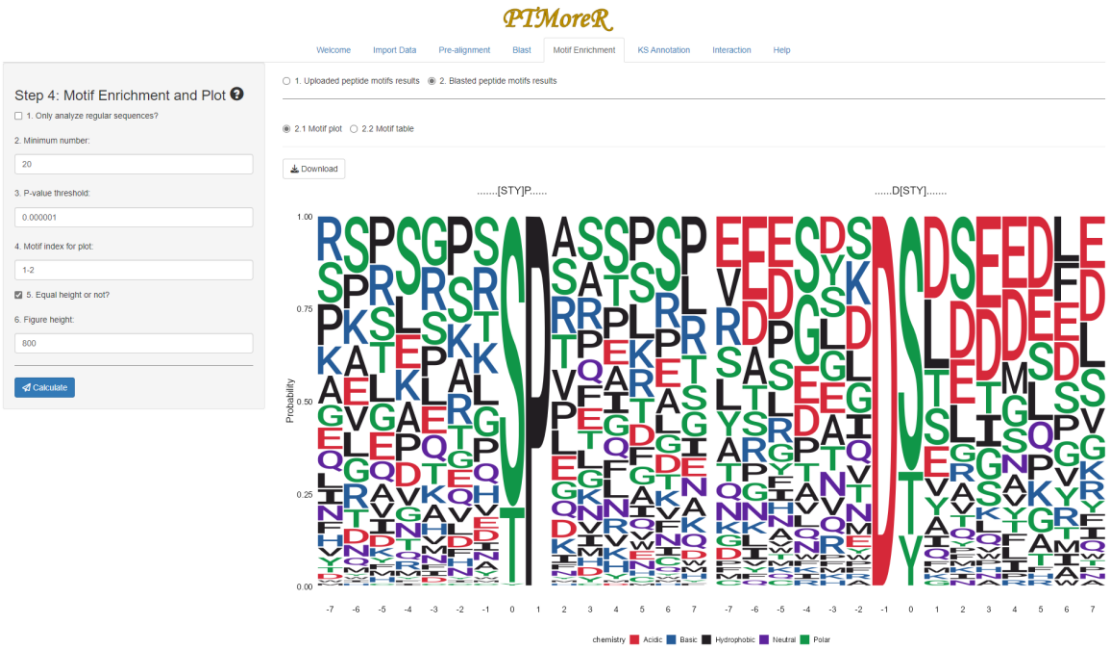
7. fold.increase: An indicator of the enrichment level of the extracted motifs. Specifically, it is calculated as (foreground matches/foreground size)/(background matches/background size).

8. Enrich.seq: those peptides are overrepresented in this motif.

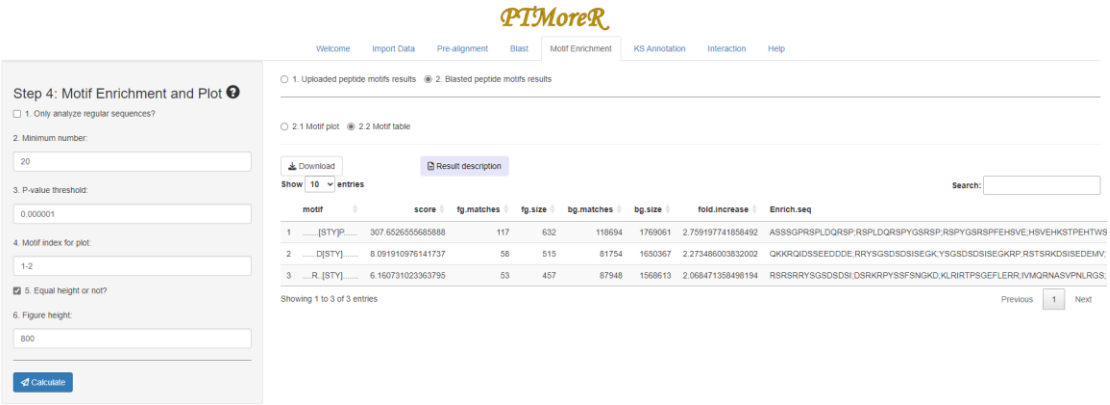
9. Enrich.pro: those proteins in which the peptides exist from Enrich.seq.

5.2.2. Blasted peptide motifs results

All results here are obtained from the blasted peptides. Similar to the above, when users click “2.1. Motif plot”, the motif plot is shown as below:



When users click “2.2. Motif table”, the motif table is shown as below:

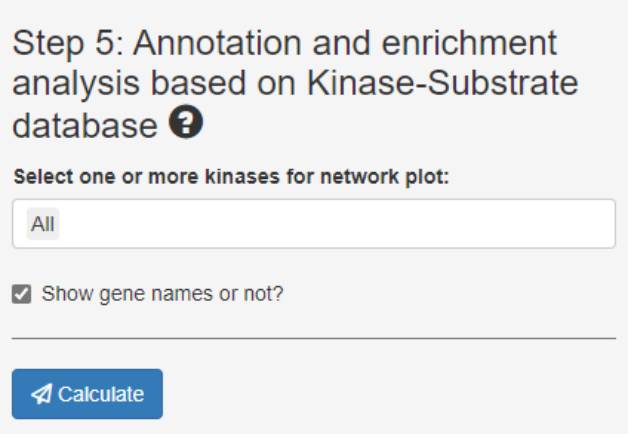


Every column means the same thing as above (Users can also check this by clicking “Result description” button).

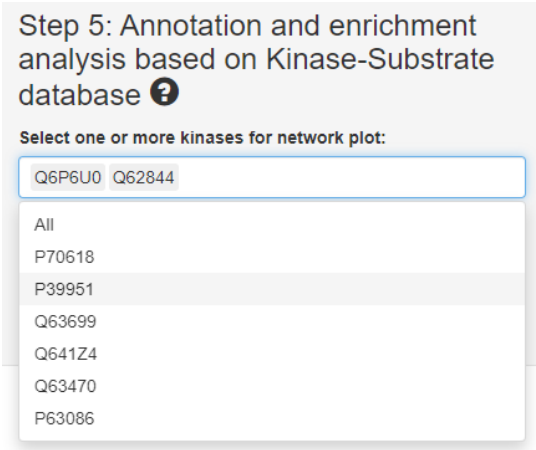
6. Annotation and enrichment analysis based on Kinase-Substrate database

This step will perform kinase-substrate annotation and enrichment for every kinase using Fisher test based on PhosphoSitePlus (9), to facilitate a site-specific analysis on phosphorylation regulation. Users should note that here is only for phoshoproteomics data, other modification data are not inappropriate here.

6.1. Parameters



Select one or more kinases for network plot: This means which kinase users want to select to show a network plot. If selecting "All", all kinases identified in users' data will be used to plot network. User can also just select one or more kinases, like below:



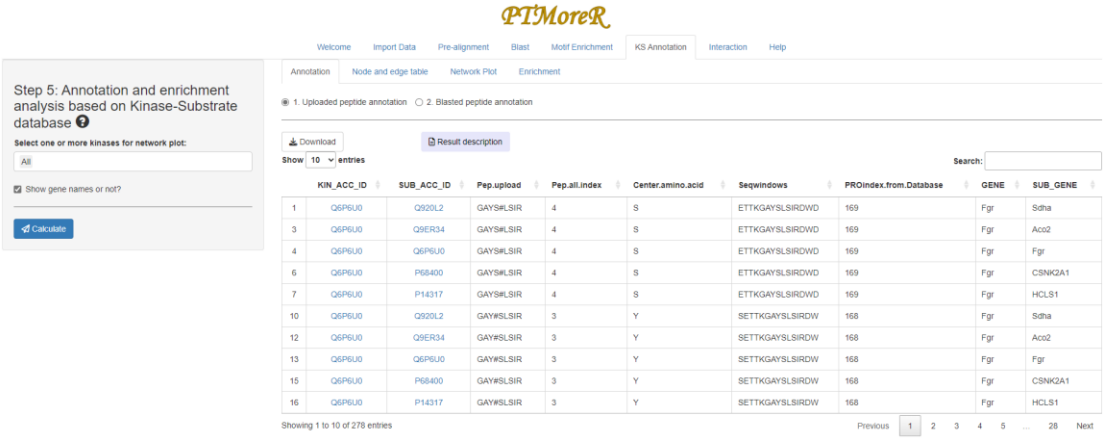
Show gene names or not? If true, the node table, the edge table, and the network plot will show the gene names, otherwise, it shows the IDs (e.g. UniProt IDs by default).

Then, users can click the "Calculate" button.

6.2. Results

6.2.1. Annotation

Here shows the annotation tables for the uploaded peptides (*1. Uploaded peptide annotation*) and the blasted peptides (*2. Blasted peptide annotation*).



	KIN_ACC_ID	SUB_ACC_ID	Pep.upload	Pep.all.index	Center.amino.acid	Seqwindows	PROindex.from.Database	GENE	SUB_GENE
1	Q6P6U0	Q920L2	GAYS#LSIR	4	S	ETTKGAYSLSIRDWD	169	Fgr	Sdha
3	Q6P6U0	Q9ER34	GAYS#LSIR	4	S	ETTKGAYSLSIRDWD	169	Fgr	Aco2
4	Q6P6U0	Q6P6U0	GAYS#LSIR	4	S	ETTKGAYSLSIRDWD	169	Fgr	Fgr
6	Q6P6U0	P68400	GAYS#LSIR	4	S	ETTKGAYSLSIRDWD	169	Fgr	CSNK2A1
7	Q6P6U0	P14317	GAYS#LSIR	4	S	ETTKGAYSLSIRDWD	169	Fgr	HCLS1
10	Q6P6U0	Q920L2	GAY#SLSIR	3	Y	SETTKGAYSLSIRDW	168	Fgr	Sdha
12	Q6P6U0	Q9ER34	GAY#SLSIR	3	Y	SETTKGAYSLSIRDW	168	Fgr	Aco2
13	Q6P6U0	Q6P6U0	GAY#SLSIR	3	Y	SETTKGAYSLSIRDW	168	Fgr	Fgr
15	Q6P6U0	P68400	GAY#SLSIR	3	Y	SETTKGAYSLSIRDW	168	Fgr	CSNK2A1
16	Q6P6U0	P14317	GAY#SLSIR	3	Y	SETTKGAYSLSIRDW	168	Fgr	HCLS1

Every column in this table means (Users can also check this by clicking “Result description” button):

1. KIN_ACC_ID: kinase uniprot id.

2. SUB_ACC_ID: substrate uniprot id.

3. Pep.upload: the original peptide.

4. Pep.all.index: the position of all modified amino acid in the peptide.

5. Center.amino.acid: the central amino acid in the aligned peptide. Or, Center.amino.acids.Other: the central amino acid mapped from the human peptides.

6. Seqwindows: the aligned standard peptides. Or, Seqwindows.Other: the standard peptides mapped from the human peptides.

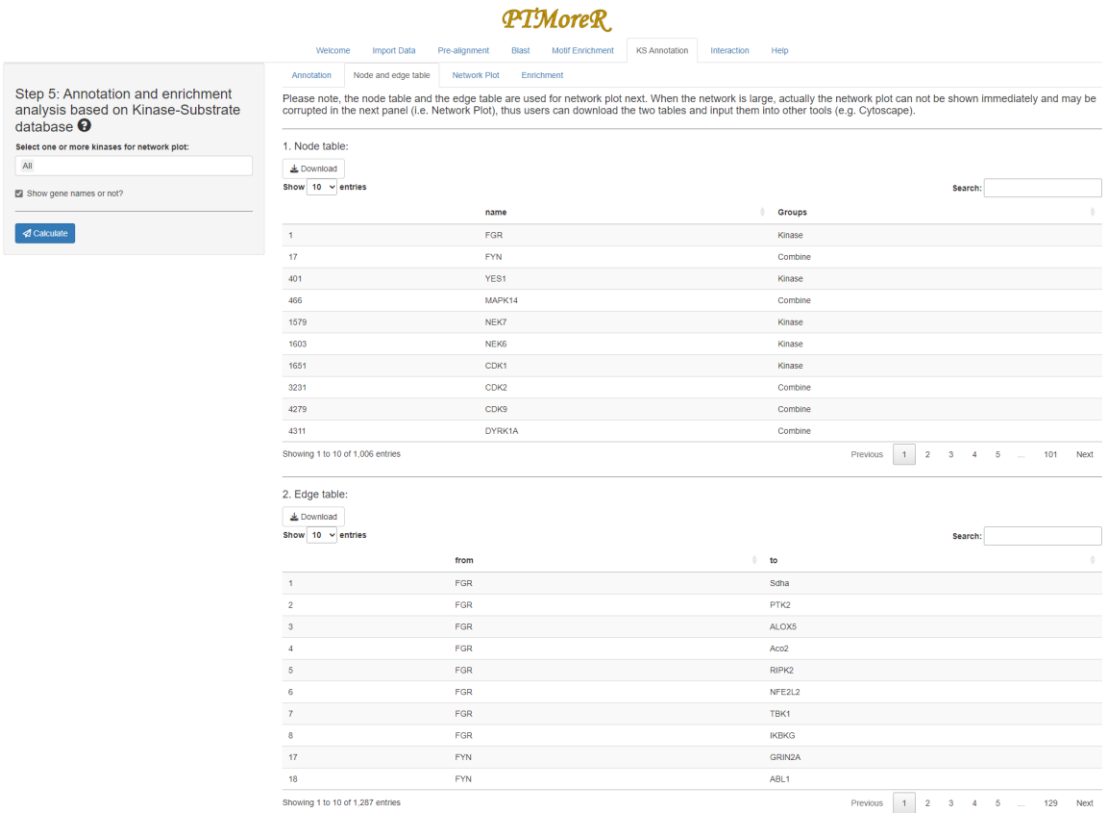
7. PROindex.from.Database: the position of modified amino acid in the protein sequence. Or, PROindex.from.Other: the position of modified amino acid in the mapped human protein sequence.

8. GENE: kinase gene name.

9. SUB_GENE: substrate gene name.

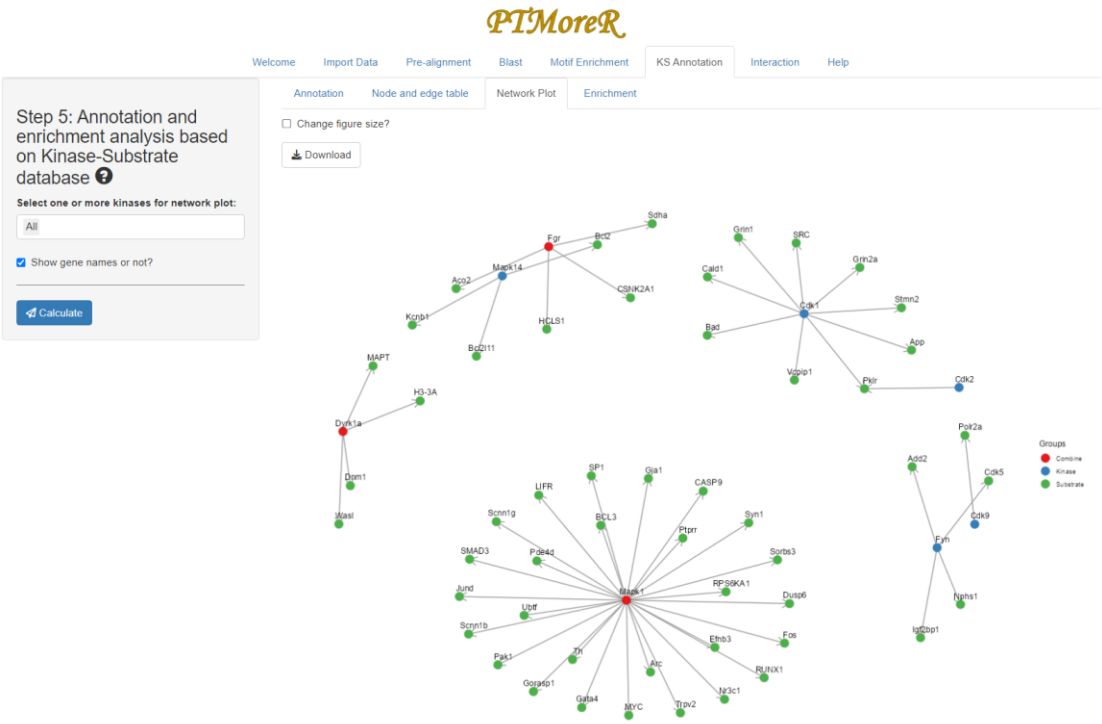
6.2.2. Node and edge table

The node table and the edge table are used for network plot next. When the network is large, actually the network plot cannot be shown immediately and may be corrupted in the next panel (i.e. Network Plot), thus users can download the two tables and input them into other tools (e.g. Cytoscape (10)). In addition, the “Combine” means one protein is a kinase and a substrate at the same time.



6.3. Network Plot

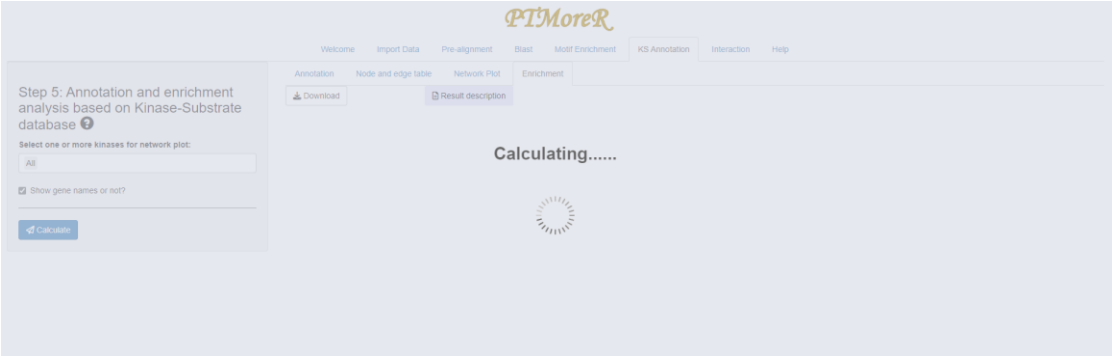
The plot shows the relationships between the kinases and the substrates obtained from above. This function is the same with motifeR in single species.



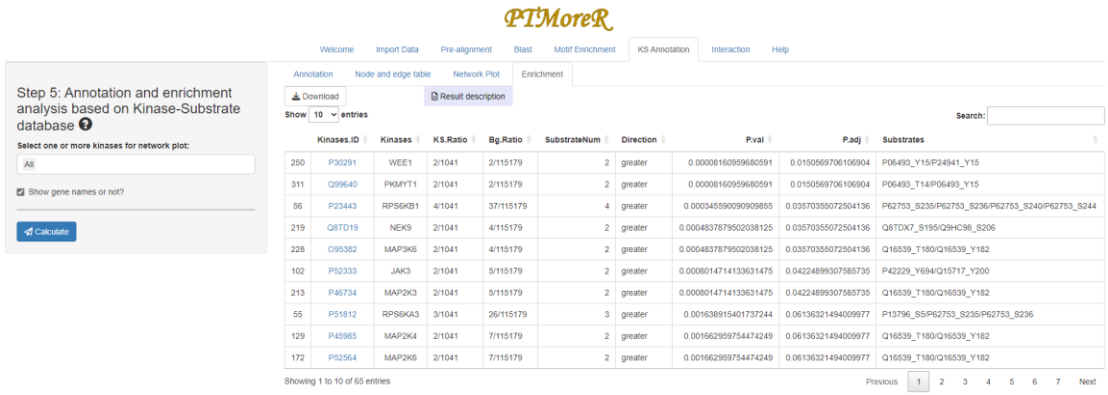
Users can click “Download” button to save this plot as a .pdf file.

6.4. Enrichment

When users click “Enrichment”, this function will process the enrichment analysis at phosphosite levels for every kinases:



Then, the results are shown as below:



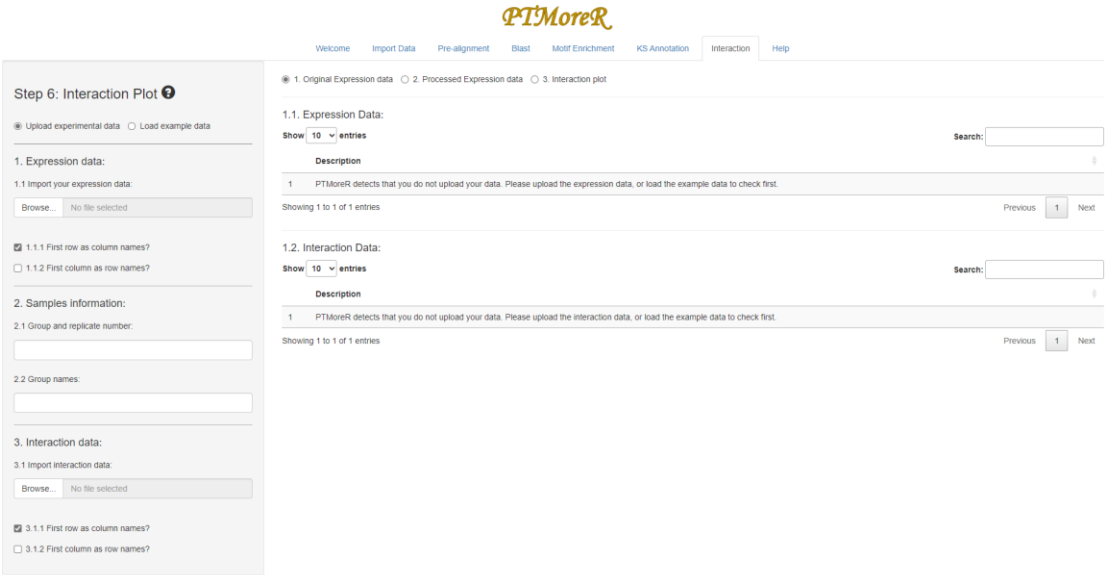
	Kinases.ID	Kinases	KS.Ratio	Bg.Ratio	SubstrateNum	Direction	P.val	P.adj	Substrates
250	P30291	WEE1	2/1041	2/115179	2	greater	0.0000816095968059 1	0.0150569706106904	P06493_Y15/P24941_Y15
311	Q99640	PKMYT1	2/1041	2/115179	2	greater	0.0000816095968059 1	0.0150569706106904	P06493_T14/P06493_Y15
56	P23443	RPS6KB1	4/1041	37/115179	4	greater	0.0003455900909098 55	0.0357035507250413 6	P62753_S235/P62753_S236/P62753_S240/P62753_S244
219	Q8TD19	NEK9	2/1041	4/115179	2	greater	0.0004837879502038125	0.0357035507250413 6	Q8TDX7_S195/Q9HC98_S206
228	O95382	MAP3K6	2/1041	4/115179	2	greater	0.0004837879502038125	0.0357035507250413 6	Q16539_T180/Q16539_Y182
102	P52333	JAK3	2/1041	5/115179	2	greater	0.0008014714133631475	0.0422489930758573 5	P42229_Y694/Q15717_Y200
213	P46734	MAP2K3	2/1041	5/115179	2	greater	0.0008014714133631475	0.0422489930758573 5	Q16539_T180/Q16539_Y182
55	P51812	RPS6KA3	3/1041	26/115179	3	greater	0.0016389154017372 44	0.0613632149400997 7	P13796_S5/P62753_S235/P62753_S236
129	P45985	MAP2K4	2/1041	7/115179	2	greater	0.0016629595754474249	0.0613632149400997 7	Q16539_T180/Q16539_Y182
172	P52564	MAP2K6	2/1041	7/115179	2	greater	0.0016629595754474249	0.0613632149400997 7	Q16539_T180/Q16539_Y182

Every column in this table means (Users can also check this by clicking “Result description” button):

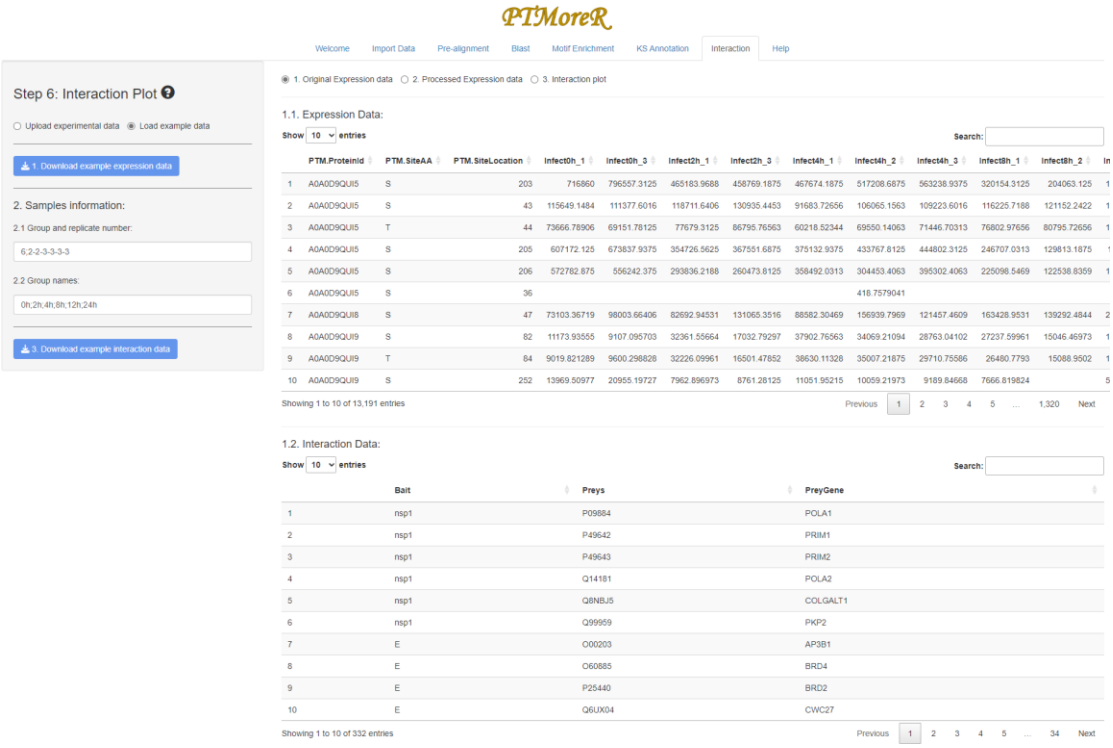
1. *Kinases.ID*: Kinase uniprot ids.
2. *Kinases*: Kinase gene names.
3. *KS.Ratio*: k/n, k means the overlap between phosphosites-of-interest and the uploaded phosphosite set, n means the number of all unique phosphosites-of-interest.
4. *Bg.Ratio*: M/N, M means the number of substrate phosphosites of each kinase in the whole phosphosite set, N means the number of phosphosites in the whole phosphosite set.
5. *SubstrateNum*: Same as k.
6. *Direction*: If KS.Ratio >= Bg.Ratio, the value is 'greater', otherwise, 'less'.
7. *P.val*: Original P value obtained from Fisher test.
8. *P.adj*: Adjusted P value based on the BH method.
9. *Substrates*: Substrate phosphosites-of-interest.

7. Interaction Plot

In some cases, it may not be convenient for the users to display a network diagram of protein-protein-modification sites relationships. This step mainly shows the plot of the interaction between the uploaded and the blasted peptides/proteins. In this step, users need to upload two kinds of data (Expression data and Interaction data) and type in the samples information, shown as below:



Users can click “Load example data”. And shown in the “*1. Original Expression data*” part below, the example data are from the African green monkey (*Chlorocebus sabaeus*) cell phosphoproteome quantified at 6 time points after SARS-CoV-2 infection (0, 2, 4, 8, 12, or 24 h) using Spectronaut (18.0.230605.50606) (11).



7.1. Data preparation

Here users should prepare two kinds of data: The modification site expression data and the interaction data.

7.1.1. The modification site expression data

Users should note that the modification site expression data should be obtained from the same species as processed in the previous steps. As shown in the example data below, the first three columns are protein ids (e.g. PTM.ProteinId), modification site amino acid (e.g. PTM.SiteAA), modification site position (e.g. PTM.SiteLocation). And the other columns are samples (e.g. The African green monkey cell samples with 6 time points after SARS-CoV-2 infection (0, 2, 4, 8, 12, or 24 h)). These data can be extracted from many popular software, such as MaxQuant, Spectronaut etc., and saved in a .csv file. The missing values are NA and shown as blank after uploaded into *PTMoreR*.

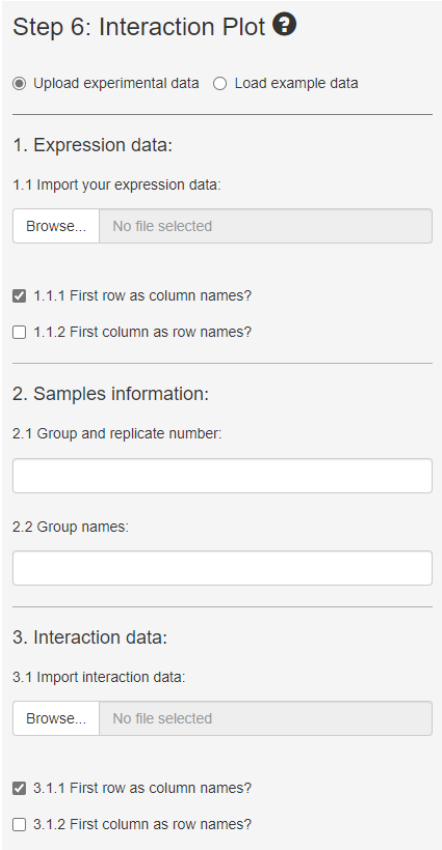
PTM.ProteinId	PTM.SiteAA	PTM.SiteLocation	Infect0h_1	Infect0h_3	Infect2h_1	Infect2h_3	Infect4h_1	Infect4h_2	Infect4h_3
A0A0D9QUI5	S	203	716860	796557.3125	465183.9688	458769.1875	467674.1875	517208.6875	563238.9375
A0A0D9QUI5	S	43	115649.1484	111377.6016	118711.6406	130935.4453	91683.72656	106065.1563	109223.6016
A0A0D9QUI5	T	44	73666.78906	69151.78125	77679.3125	86795.76563	60218.52344	69550.14063	71446.70313
A0A0D9QUI5	S	205	607172.125	673837.9375	354726.5625	367551.6875	375132.9375	433767.8125	444802.3125
A0A0D9QUI5	S	206	572782.875	556242.375	293836.2188	260473.8125	358492.0313	304453.4063	395302.4063
A0A0D9QUI5	S	36						418.7579041	
A0A0D9QUI8	S	47	73103.36719	98003.66406	82692.94531	131065.3516	88582.30469	156939.7969	121457.4609
A0A0D9QUI9	S	82	11173.93555	9107.095703	32361.55664	17032.79297	37902.76563	34069.21094	28763.04102
A0A0D9QUI9	T	84	9019.821289	9600.298828	32226.09961	16501.47852	38630.11328	35007.21875	29710.75586
A0A0D9QUI9	S	252	13969.50977	20955.19727	7962.896973	8761.28125	11051.95215	10059.21973	9189.84668

7.1.2. The interaction data

Users should know the interaction data in advance, which could be protein-protein interaction data, or drug-protein interaction data and so on. Three are three columns in this table: The first one is protein ids/names from one species (e.g. SARS-CoV-2 protein names), or drug names; the second one is protein ids from the species that users blast to in the previous steps (e.g. human protein ids, as users blast the phosphopeptides from green monkey to human); the third one is gene names relative to the protein ids in the second column.

Bait	Preys	PreyGene
nsp1	P09884	POLA1
nsp1	P49642	PRIM1
nsp1	P49643	PRIM2
nsp1	Q14181	POLA2
nsp1	Q8NBJ5	COLGALT1
nsp1	Q99959	PKP2
E	O00203	AP3B1
E	O60885	BRD4
E	P25440	BRD2
E	Q6UX04	CWC27

7.2. Parameters



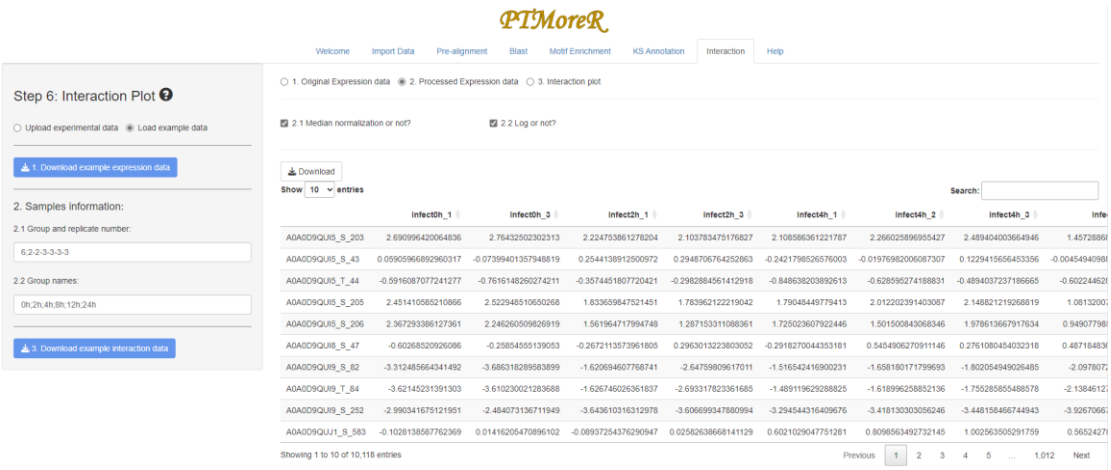
The parameters in the "*1. Expression data*" and the "*3. Interaction data*" parts are similar and easy. Users just click the "*Browse*" and find their data in a .csv file, then notice that whether the first row/column is used as row/column names, if true, they should select relative parameters.

For the "*2. Samples information*" part, the "*2.1. Group and replicate number*" means users can type in the group number and the biological replicate number here. Please note, the group number and replicate number are linked with ";", and the replicate number of each group is linked with "-". For example, if you have two groups, each group has three replicates, then you should type in "2;3-3" here. Similarly, if you have 3 groups with 5 replicates in every groups, you should type in "3;5-5-5".

2.2 Group names: Type in the group names of your samples. Please note, the group names are linked with ";". For example, there are 6 time points after SARS-CoV-2 infection (0, 2, 4, 8, 12, or 24 h), you can type in "0h;2h;4h;8h;12h;24h".

7.3. Processed Expression data

After uploading the proper data and setting up the right parameters, users can click “2. Processed Expression data” (see below), and this tool will process the data automatically for users. By default, those sites with over 50% missing ratios across all samples were removed and missing values were imputed with the k-Nearest Neighbor algorithm provided in NAguideR (12).

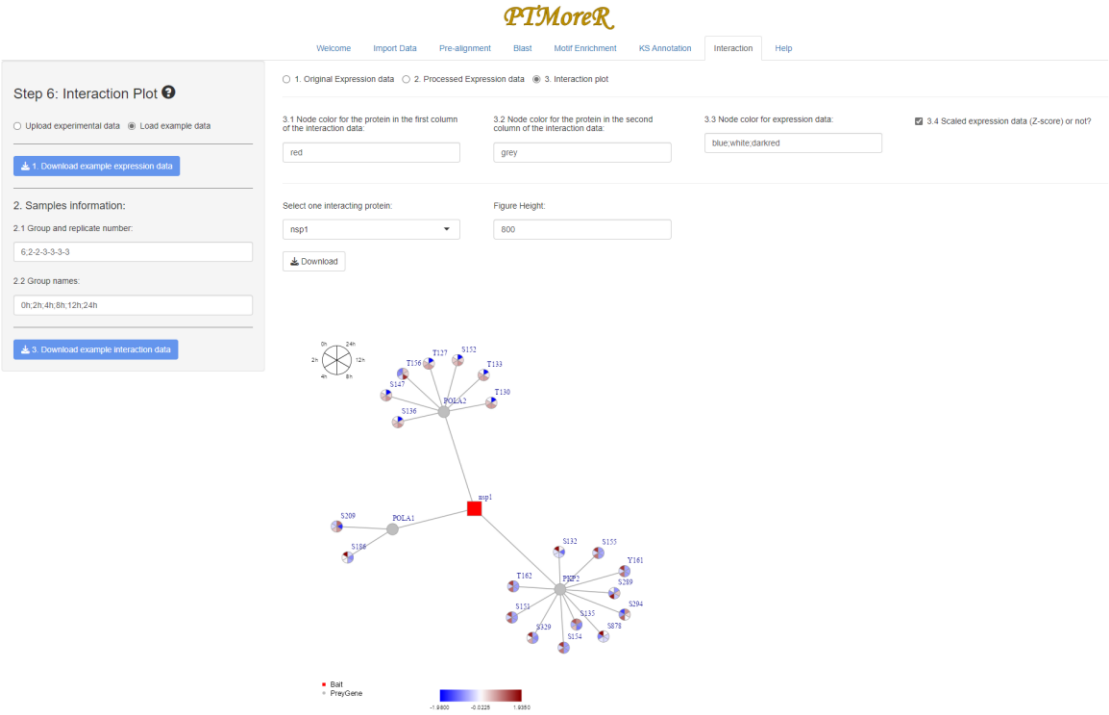


2.1. Median normalization or not: if true, *PTMoreR* will process median normalization for original data. (Note, *PTMoreR* was not designed to perform sophisticated normalization analysis. Any normalized datasets with NA can be accepted for analysis).

2.2. Log or not: if true, the data will be transformed to the logarithmic scale with base 2.

7.4. Interaction plot

In this step, PTMoreR will show the final interaction plot based on the uploaded expression data and the interaction data.



3.1 Node color for the protein in the first column of the interaction data: Three are three columns in the interaction data (see "*7.1.2. The interaction data*" part), the node color for one protein in the first column (e.g. One SARS-CoV-2 protein in the *Bait* column).

3.2 Node color for the protein in the second column of the interaction data: Similar as above, the node color for one protein in the second column (e.g. One human protein in the *Preys* column).

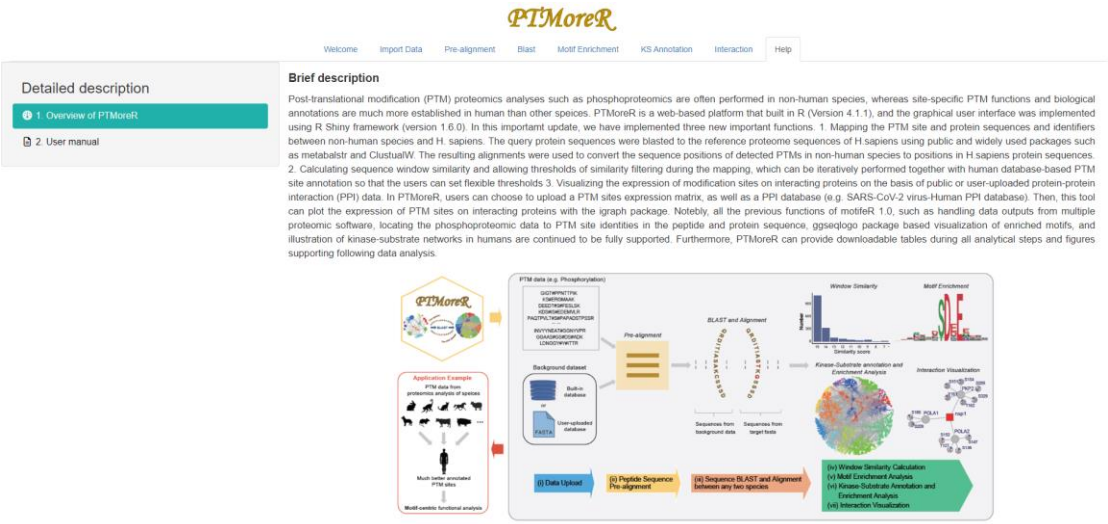
3.3 Node color for expression data: Three color names which are linked with ";" for the expression data (e.g. blue;white;darkred). The first color (e.g. blue) indicates the lowest expression value, the second one (e.g. white) indicates the middle expression value, and the third one (e.g. darkred) indicates the largest expression value.

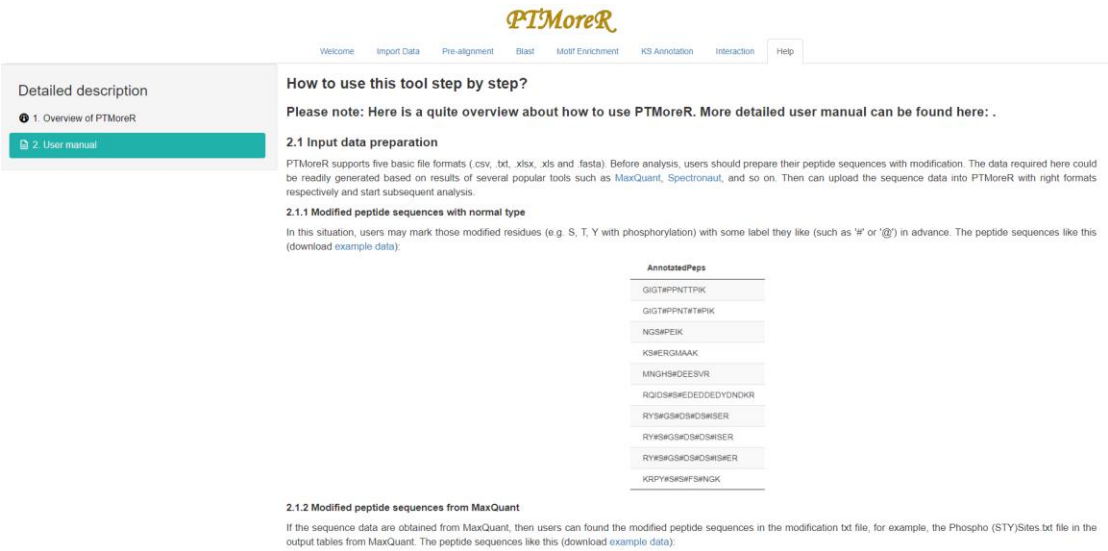
3.4 Scaled expression data (Z-score) or not? If true, the expression data will be scaled by rows, which means the mean of all of the values in each row is 0 and the standard deviation is 1.

3.5 Select one interacting protein: Users can change this parameter to select any protein in the first column (e.g. One SARS-CoV-2 protein in the *Bait* column) and the plot will show the selected protein.

8. Help

This part provides online introductions and the detailed operation manual about *PTMoreR*.





9. How to run this tool locally?

PTMoreR is an open source software for non-commercial use and all codes can be obtained on our GitHub: https://github.com/wangshisheng/PTMoreR. If users want to run *PTMoreR* on their own computer independent of the internet speed, they should operate as below:

As this tool was developed with R, you may:

a) Install R. You can download R from here: https://www.r-project.org/.

b) Install RStudio. (Recommendatory but not necessary). You can download RStudio from here: https://www.rstudio.com/.

c) Check packages. After installing R and RStudio, you should check whether you have installed these packages (shiny, shinyjs, shinyWidgets, shinyBS, DT, data.table, openxlsx, Biostrings, GenomicFeatures, rtracklayer, stringi, stringr, ggsci, ggplot2, ggrepel, msa, tidyr, ggraph, graphlayouts, scales, impute, igraph, scatterpie, plotfunctions, mapplots, devtools). You may run the codes below to check them:

```
if(!require(pacman)) install.packages("pacman")
pacman::p_load(shiny, shinyjs, shinyWidgets, shinyBS, DT, data.table, openxlsx, Biostrings, GenomicFeatures, rtracklayer, stringi, stringr, ggsci, ggplot2, ggrepel, msa, tidyr, ggraph, graphlayouts, scales, impute, igraph, scatterpie, plotfunctions, mapplots, devtools)
```

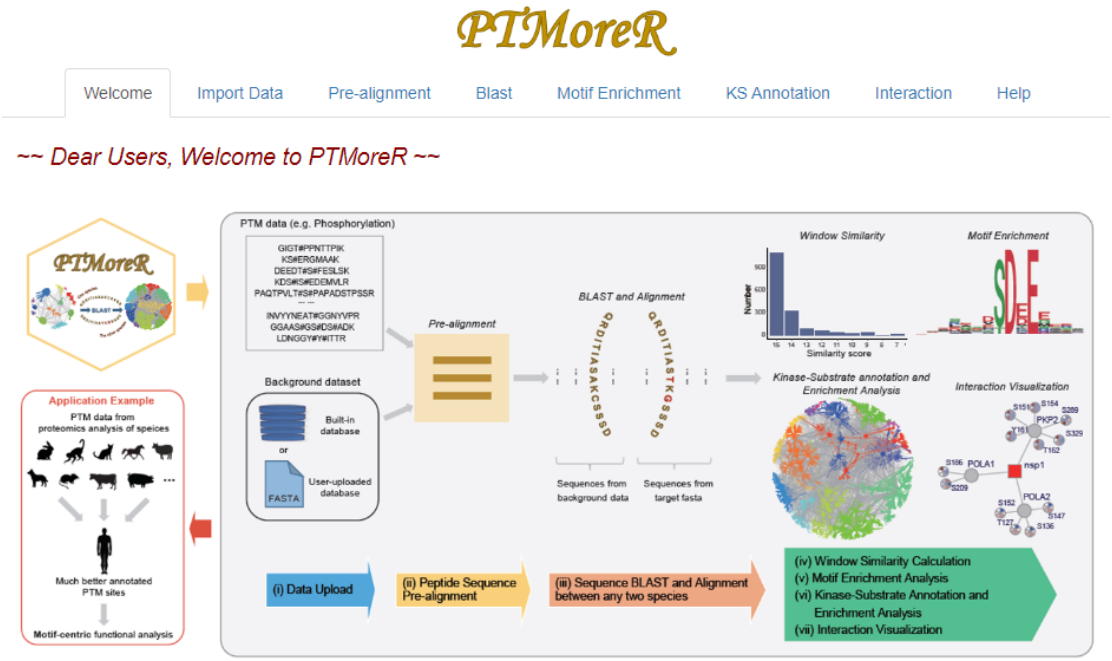
Then install some packages from GitHub, as below:

```
devtools::install_github("drostlab/metablastr", build_vignettes = TRUE, dependencies = TRUE)
devtools::install_github('omarwagih/rmotifx')
devtools::install_github("omarwagih/ggseqlogo")
```

d) Run this tool locally

```
if(!require(PTMoreR)) devtools::install_github("wangshisheng/PTMoreR")
library(PTMoreR)
PTMoreR_app()
```

Then PTMoreR will be started as below, and the detailed operation about PTMoreR can be found in the Supplementary Notes part 1-8 above.



PTMoreR is a web-based tool, which possesses the core functions, including:

1. Mapping the PTM site and protein sequences and identifiers between non-human species and H. sapiens;
2. Calculating sequence window similarity and allowing thresholds of similarity filtering for motif mapping;
3. Offering more flexible annotation based on kinase-substrate databases (e.g. PhosphoSitePlus) and network plots;
4. Visualizing the expression of modification sites on interacting proteins on the basis of public or user-uploaded protein-protein interaction data.

In addition, this tool supports both online access and local installation. The source codes and installation instructions can be available in the GitHub repository: https://github.com/wangshisheng/PTMoreR under an MIT license.

Finally, PTMoreR is developed by R shiny (Version 1.6.0), and is free and open to all users with no login requirement. It can be readily accessed by all popular web browsers including Google Chrome, Mozilla Firefox, Safari and Internet Explorer 10 (or later), and so on. We would highly appreciate that if you could send your feedback about any bug or feature request to Shisheng Wang at shishengwang@wchscu.cn.

*^_^ Enjoy yourself in PTMoreR ^_^*

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III. References

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